CS6650 Assignment 3 — ChatFlow

Github url: https://github.com/wang-yanyue/cs6650-assignment3

Database Design Document

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2.1 Database Choice: PostgreSQL 15

PostgreSQL was selected primarily for three reasons specific to this assignment:

1. **Native materialized views** — required for pre-computing analytics queries (A1-A4) without an external caching layer
2. **ON CONFLICT DO NOTHING** — enables idempotent upserts, satisfying the deduplication requirement at the DB level with no application logic needed
3. **TIMESTAMPTZ** — timezone-aware timestamps essential for distributed message ordering across servers

MySQL was ruled out due to lack of materialized views. MongoDB was ruled out because the query patterns (time-range scans, aggregations, joins) favor relational structure.

2.2 Schema Design

Two tables and five materialized views:

messages — core storage. message_id (UUID) is the primary key, enforcing uniqueness. created_at (TIMESTAMPTZ) drives all time-range queries.

messages_dlq — dead letter queue for failed writes. Adds error_reason and failed_at columns for diagnostics and manual replay.

Materialized views: mv_msg_per_minute, mv_msg_per_second (A1), mv_top_users (A2), mv_top_rooms (A3), mv_user_participation (A4). Each carries a UNIQUE index to support REFRESH CONCURRENTLY.

2.3 Indexing Strategy

Three indexes cover all four core queries:

- **idx_room_time (room_id, created_at DESC)** — Q1 room+time range scans

- **idx_user_time (user_id, created_at DESC)** — Q2 user history and Q4 room participation
- **idx_time (created_at DESC)** — Q3 pure time-window distinct user counts

Write overhead is 4 B-tree updates per insert (3 indexes + PK). This is acceptable because batch inserts (1,000 rows/batch) amortize the cost — sustained throughput reached **3,960 msg/s** with no index bottlenecks observed.

2.4 Scaling Considerations

At current load the single PostgreSQL instance is sufficient — 1M messages were written with 99.92% buffer hit rate and zero deadlocks. For larger scale:

- **Read replicas** — analytics queries hit materialized views and can be routed to replicas, keeping writes on the primary
- **Table partitioning** — range partition messages by created_at monthly when row count exceeds ~100M
- **HikariCP** — pool sized at 20 max connections, 5 minimum idle, matching the 4 DB writer threads with headroom for API query threads

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2.5 Backup and Recovery

- **Backups**: Daily pg_dump compressed to S3 with 7-day retention. WAL archiving enables point-in-time recovery within any 5-minute window.
- **Message loss protection**: The 500,000-capacity write-behind buffer provides ~125 seconds of protection at peak load before data loss. The circuit breaker (opens after 5 failures, 30s cooldown) routes messages to messages_dlq during outages for manual replay.
- **Idempotent recovery**: ON CONFLICT (message_id) DO NOTHING ensures DLQ replay produces no duplicates.

Performance Report

1. Write Performance

1.1 Maximum Sustained Write Throughput

The system was tested across three progressive load levels using batchSize=1000, flushIntervalMs=500, and 4 DatabaseWriter threads:

Test	Messages	Duration	Throughput	Final DB Count
Baseline	500,000	127.7s	3,914 msg/s	500,000 (100%)
Stress	1,000,000	252.5s	3,960 msg/s	1,000,000 (100%)
Endurance	3,123,900	30 min	1,735 msg/s	3,123,900 (100%)

Maximum sustained throughput: **3,960 msg/s** with zero message loss across all tests. The endurance test ran at a lower rate (1,735 msg/s) due to the Python publisher's sleep overhead, but demonstrated stable throughput with no degradation over 30 minutes.

1.2 Latency Percentiles

Direct write latency was not instrumented at the application level. However, the write-behind architecture means end-to-end latency has two components:

- **RabbitMQ → buffer**: near-zero (in-memory enqueue)
- **Buffer → PostgreSQL**: up to flushIntervalMs (500ms) + batch execution time

At 3,960 msg/s with batchSize=1000 and 4 writer threads, each batch takes approximately 25ms to execute (estimated from throughput). This gives approximate percentiles:

Percentile	Estimated Latency
p50	~250ms (mid-flush-interval)
p95	~480ms (near flush deadline)
p99	~520ms (flush + batch execution)

These are write-behind latencies — RabbitMQ acknowledgment happens immediately on receipt, so the real-time chat experience is unaffected.

1.3 Batch Size Optimization Results

Five configurations were tested with 50,000 messages each:

Test	batchSize	flushIntervalMs	DB Count	Notes
T1	100	100	1,291	Too small — frequent small writes
T2	500	500	1,344	Improved but still suboptimal
T3	1,000	500	51,114	Optimal — best throughput/latency balance
T4	5,000	1,000	50,000	High latency, risk of buffer accumulation
T5	1,000	100	50,000	Unnecessary flush frequency, higher CPU

Optimal configuration: batchSize=1000, flushIntervalMs=500. T3 achieved the best balance — large enough batches to amortize PostgreSQL round-trip cost, short enough flush interval to bound write latency under 500ms.

Note: T1 and T2 low DB counts reflect tests run before the Python direct publisher was adopted. The server-side bottleneck at that stage limited messages reaching RabbitMQ.

1.4 Resource Utilization

At peak load (3,960 msg/s):

- **HikariCP connections**: 9-11 active out of 20 maximum — pool was never exhausted
- **PostgreSQL buffer hit rate**: 99.88-99.92% — virtually all reads served from shared_buffers, minimal disk I/O for reads
- **RabbitMQ queue depth**: consistently 0-7 messages — consumer drained queues as fast as messages arrived
- **Write amplification**: 4 B-tree updates per insert (3 indexes + PK), absorbed by batch writes

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2. System Stability

2.1 Queue Depth

RabbitMQ queue depth remained at 0-7 messages throughout all three load tests. This confirms the consumer's dynamic scaling and prefetch configuration (50 messages/worker × 5 workers = 250 in-flight) was sufficient to prevent queue buildup at all tested load levels. No queue overflow or message drop was observed.

2.2 Database Performance Metrics

Across all tests, PostgreSQL showed no signs of stress:

Metric	Baseline	Stress	Endurance
Buffer hit rate	99.88%	99.92%	99.29%
Deadlocks	0	0	0
Active connections (peak)	10	11	16
Total commits	379,196	680,194	1,512,706

Buffer hit rate dropped slightly during the endurance test (99.93% → 99.29%) as the working dataset grew larger than PostgreSQL's shared_buffers, requiring more disk reads. Still well above the 95% threshold considered healthy.

2.3 Memory Usage Patterns

Time	Memory Used	Free	Disk Used
Test start	442MB	570MB	60%
15 minutes	961MB	76MB	76%
30 minutes	988MB	129MB	83%

Memory stabilized at ~988MB after 15 minutes with no further growth — indicating no memory leak in the consumer JVM or PostgreSQL. The increase from 442MB to 988MB reflects PostgreSQL expanding its shared_buffers cache as the dataset grew, which is expected and desirable behavior.

2.4 Connection Pool Statistics

HikariCP performed as configured throughout all tests:

- **Minimum idle**: 5 connections maintained at startup
- **Peak active**: 16 connections during endurance test (under 20 max)

- **Post-test**: connections returned to 6-8 idle, confirming proper connection release
- **No timeouts or exhaustion events** observed in any test

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3. Bottleneck Analysis

3.1 Primary Bottlenecks Identified

1. Disk space — the most pressing constraint. Over 30 minutes at 1,735 msg/s, disk usage grew from 60% to 83% (4.5GB → 6.3GB). At this rate the 7.6GB volume would reach capacity in approximately 75 minutes of sustained load. This is driven by PostgreSQL WAL, table data, and index growth.

2. Server-side message loss — the WebSocket server (server-v2) dropped approximately 63% of messages under load due to channel pool contention (pool size=16) and circuit breaker trips. This limited real-world throughput through the full pipeline and required bypassing the server with a direct RabbitMQ publisher for load tests.

3. Single consumer instance — the write-behind pipeline runs on one EC2 instance. At 3,960 msg/s the 4 DatabaseWriter threads and HikariCP pool (20 connections) are not yet saturated, but a single point of failure exists.

3.2 Proposed Solutions

Bottleneck	Solution
Disk space	Attach EBS volume (100GB+), enable PostgreSQL table partitioning by month to allow dropping old partitions
Server message loss	Increase channelPoolSize from 16 to 64; tune circuit breaker failureThreshold from 3 to 10
Single consumer	Deploy 2 consumer instances; partition queues (room.1-10 / room.11-20) between them

3.3 Trade-offs Made

Write-behind vs. durability: Acknowledging messages to RabbitMQ before DB write maximizes throughput (3,960 msg/s achieved) but means in-flight messages in the buffer are lost on consumer crash. The DLQ and circuit breaker mitigate but do not eliminate this risk. A synchronous write approach would guarantee durability but throughput would drop to ~500 msg/s (single-row insert rate).

batchSize=1000 vs. larger: Larger batches (T4: 5,000) marginally improve raw write throughput but increase maximum write latency to ~1,000ms and risk larger buffer accumulation during a DB outage. batchSize=1000 was chosen as the safer balance.

Materialized view refresh interval (30s): Analytics data is up to 30 seconds stale. A shorter interval (5s) would improve freshness but adds refresh overhead during high write load, as each REFRESH CONCURRENTLY competes with active inserts for I/O.

Configuration Files

All configuration is defined in consumer-v3/src/main/resources/application.properties. Runtime overrides were passed as JVM arguments during load tests (e.g. --chat.db.batchSize=100 for T1).

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1. Database Connection Settings

```
# PostgreSQL 15 — deployed on same EC2 as consumer

spring.datasource.url=jdbc:postgresql://127.0.0.1:5432/chatflow

spring.datasource.username=chatuser

spring.datasource.password=chatpassword

spring.datasource.driver-class-name=org.postgresql.Driver
```

127.0.0.1 is used instead of localhost to force TCP connection, bypassing PostgreSQL's peer authentication which only works on Unix sockets.

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2. Thread Pool Configurations

```
# RabbitMQ consumer workers (CachedThreadPool)

chat.consumer.minWorkers=5       # initial worker threads at startup

chat.consumer.maxWorkers=40      # hard ceiling for dynamic scale-up

chat.consumer.scaleUpThreshold=500    # scale up when queue depth > 500
```

```
chat.consumer.scaleDownThreshold=50   # scale down when queue depth < 50
chat.consumer.scaleIntervalMs=5000    # scaler checks every 5 seconds
chat.consumer.prefetch=50             # max in-flight messages per worker

# Database writer threads (FixedThreadPool — separate from consumer pool)
chat.db.writerThreads=4          # 4 dedicated DB writer threads
chat.db.bufferCapacity=500000    # in-memory buffer between consumer and writers
```

Three thread pools run independently:

- **Consumer pool** (CachedThreadPool): 5-40 threads, drains RabbitMQ into the write buffer
- **DB writer pool** (FixedThreadPool of 4): drains write buffer into PostgreSQL in batches
- **Scheduler** (single thread): runs StatsAggregator to refresh materialized views every 30s

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3. Batch Processing Parameters

```
chat.db.batchSize=1000       # rows per batch insert (optimal from T1-T5 tests)
chat.db.flushIntervalMs=500   # max wait before flushing a partial batch
chat.db.maxRetries=3         # retry attempts on write failure
                             # backoff: 100ms → 200ms → 400ms (exponential)
chat.db.bufferCapacity=500000 # buffer holds up to 500k messages before drops

# Materialized view refresh
chat.db.statsRefreshIntervalMs=30000  # refresh analytics views every 30 seconds
```

Batch size selection rationale (from T1-T5 results):

- batchSize=100 was too small — high round-trip overhead per batch
- batchSize=5000 introduced up to 1,000ms write latency

- batchSize=1000 with flushIntervalMs=500 achieved 3,960 msg/s with ~500ms max latency

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4. HikariCP Connection Pool

spring.datasource.hikari.pool-name=ChatflowPool

spring.datasource.hikari.maximum-pool-size=20 # max concurrent DB connections

spring.datasource.hikari.minimum-idle=5 # warm connections kept ready

spring.datasource.hikari.connection-timeout=30000 # fail after 30s if no connection available

spring.datasource.hikari.idle-timeout=600000 # close idle connections after 10 min

spring.datasource.hikari.max-lifetime=1800000 # recycle connections after 30 min

Pool sizing rationale: 4 DB writer threads + up to 16 concurrent API query threads = 20 maximum. Minimum idle of 5 prevents cold-start latency on the first API request after startup.

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5. Circuit Breaker Thresholds

chat.db.circuitBreaker.failureThreshold=5 # open after 5 consecutive batch failures

chat.db.circuitBreaker.cooldownMillis=30000 # stay open for 30 seconds before retry

State transitions:

- **CLOSED** (normal): all batch writes proceed to PostgreSQL
- **OPEN** (DB down): after 5 consecutive failures, circuit opens. Incoming messages are routed to messages_dlq for 30 seconds
- **HALF-OPEN** (recovery trial): one trial batch is attempted after 30s cooldown. Success → CLOSED; failure → OPEN again

Threshold rationale: failureThreshold=5 with exponential backoff (100ms → 400ms) means the circuit opens after ~800ms of failures — fast enough to protect the buffer from filling, slow enough to avoid false trips on transient PostgreSQL hiccups.

Appendix: Metrics API & Results Log

Part 1 Screenshots (Client Logs):

```
PS C:\Users\yanyu\OneDrive\Desktop\2026 Spring\CS 6650 Building Scalable Distributed Systems\Assignment 3\client> java -jar "target\client-0.0.1-SNAPSHOT.jar" `ws://server-v2-alb-646704798.us-west-2.elb.amazon
aws.com/chat `500000 `http://44.249.100.29:8082/api/metrics
Starting load test against: ws://server-v2-alb-646704798.us-west-2.elb.amazonaws.com/chat
Total messages: 500000
Warmup phase complete.
Warmup messages sent:     32000
Warmup duration (s):      0.834
Warmup throughput (msg/s): 38346.78

=== Test Summary ===
Successful messages:      500000
Failed messages:          0
Total runtime (s):        6.239
Overall throughput (msg/s): 80143.36
Total connections opened:  96
Total reconnects:         0

=== Querying Metrics API ===
URL: http://44.249.100.29:8082/api/metrics
Sending requestâ??
HTTP status: 200

=== Metrics API Response ===
{
  "generatedAt": "2026-03-29T08:55:37.006571183Z",
  "totalMessagesInDB": 3297,
  "queryWindowStart": "2026-03-29T08:49:36.703735Z",
  "queryWindowEnd": "2026-03-29T08:55:30.937919Z",
  "queriedRoomId": "1",
  "queriedUserId": "19763",
  "coreQueries": {
    "Q1_roomMessagesInTimeRange": {
      "description": "Get messages for a room in time range",
      "performanceTarget": "< 100 ms for 1 000 messages",
      "roomId": "1",
      "startTime": "2026-03-29T08:49:36.703735Z",
      "endTime": "2026-03-29T08:55:30.937919Z",
      "messageCount": 155,
      "sampleMessages": [
        {
          "message_id": "2409c722-f262-4a10-98ac-5951d2a6afb0",
          "room_id": "1",
          "user_id": "54316",
```

```
    "username": "user54316",
    "message": "Sample message 37 - 31f919cc-8f86-415c-b61e-b1a3b1d1a259",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.780Z"
  },
  {
    "message_id": "8316d7d4-3e6a-4fdf-9174-4080ac89bb2b",
    "room_id": "1",
    "user_id": "8955",
    "username": "user8955",
    "message": "Sample message 21 - 54f0f388-905d-4828-b4a3-d2f54fe3dd5c",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.780Z"
  },
  {
    "message_id": "ccfcafd3-f2f1-47ee-8761-8f7cf443f9c7",
    "room_id": "1",
    "user_id": "95291",
    "username": "user95291",
    "message": "Sample message 7 - c4efcde4-c8d8-4df7-88f9-272cda44216a",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.782Z"
  },
  {
    "message_id": "64b4a491-697d-45a2-8b6a-cf2352f26509",
    "room_id": "1",
    "user_id": "969",
    "username": "user969",
    "message": "Sample message 17 - 141f4d0c-0a8f-4706-b10f-fa5fc2aa42fb",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.782Z"
  },
  {
    "message_id": "99ffa5ac-7c0e-4e8f-b171-75fed873bc2c",
    "room_id": "1",
    "user_id": "59377",
    "username": "user59377",
    "message": "Sample message 45 - ca81dca5-499c-467e-a24c-727776911924",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.783Z"
  },
```

```
},
{
  "message_id": "fa631e7e-8bcc-4ad2-9289-f25489cfcee8",
  "room_id": "1",
  "user_id": "54671",
  "username": "user54671",
  "message": "Sample message 13 - 6e93ec7a-3c5d-407c-aca6-8a26b1310b77",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.35.89",
  "created_at": "2026-03-29T08:49:36.784Z"
},
{
  "message_id": "f5c756af-1116-4eac-a967-30bef4aebcfa",
  "room_id": "1",
  "user_id": "11044",
  "username": "user11044",
  "message": "Sample message 22 - 40027184-9c73-4ddc-b4d9-d87c3231e204",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.35.89",
  "created_at": "2026-03-29T08:49:36.784Z"
},
{
  "message_id": "d643ba62-59e6-4b9c-b3b8-9956416c6605",
  "room_id": "1",
  "user_id": "81520",
  "username": "user81520",
  "message": "Sample message 45 - ca81dca5-499c-467e-a24c-727776911924",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.35.89",
  "created_at": "2026-03-29T08:49:36.785Z"
},
{
  "message_id": "84d5bbe5-bd54-4d92-8ada-5d17a7918117",
  "room_id": "1",
  "user_id": "59734",
  "username": "user59734",
  "message": "Sample message 12 - 0eb691da-ceaa-4b3a-a990-1db6653a85db",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.35.89",
  "created_at": "2026-03-29T08:49:36.785Z"
},
{
  "message_id": "032bc900-0a26-4357-b69c-c89dd55ecfe8",
  "room_id": "1",
  "user_id": "82331",
```

```
    "username": "user82331",
    "message": "Sample message 12 - 0eb691da-ceaa-4b3a-a990-1db6653a85db",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.786Z"
  },
  {
    "message_id": "060debf6-9916-4283-9bf1-b13a27293d78",
    "room_id": "1",
    "user_id": "49267",
    "username": "user49267",
    "message": "Sample message 1 - a29ad57c-14db-4406-9459-e3bbaf327399",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.786Z"
  },
  {
    "message_id": "8894e022-48c4-4cee-b58d-83041dae1e8f",
    "room_id": "1",
    "user_id": "39952",
    "username": "user39952",
    "message": "Sample message 46 - 1adfc3ea-3c13-4989-96ee-6d0c13d712cb",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.787Z"
  },
  {
    "message_id": "3eea61de-9fec-4c0d-a89f-9eba3aeb1e99",
    "room_id": "1",
    "user_id": "94273",
    "username": "user94273",
    "message": "Sample message 20 - 4d4b9c39-494f-450d-a009-7654d7cdcecd",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.788Z"
  },
  {
    "message_id": "aa79ec85-5876-429e-be87-aa8a018c4109",
    "room_id": "1",
    "user_id": "73589",
    "username": "user73589",
    "message": "Sample message 8 - 0fffdaf4-6060-46c3-ba3a-9ad647851459",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.788Z"
  },
```

```
  {
    "message_id": "b40875d4-e751-4f88-bb31-7a82be3e8e57",
    "room_id": "1",
    "user_id": "94417",
    "username": "user94417",
    "message": "Sample message 30 - fe2e39bb-0026-4e07-8305-947c39912346",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.789Z"
  },
  {
    "message_id": "7c001953-0f6f-48e8-bfcd-35a064f5b1aa",
    "room_id": "1",
    "user_id": "15283",
    "username": "user15283",
    "message": "Sample message 11 - 8cb34af7-cc6e-4167-aafa-a748829883af",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.789Z"
  },
  {
    "message_id": "74ec1d04-4203-4c56-9a0a-6b122f934dc1",
    "room_id": "1",
    "user_id": "23870",
    "username": "user23870",
    "message": "Sample message 3 - 85d77f26-5481-4ec2-8263-7eadda29954d",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.792Z"
  },
  {
    "message_id": "92ff72de-1834-4935-9837-bc8b79f9befa",
    "room_id": "1",
    "user_id": "85940",
    "username": "user85940",
    "message": "Sample message 16 - 0befb98a-4cff-4e63-8f20-57b22c82bf53",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.793Z"
  },
  {
    "message_id": "04f563c7-e095-4b12-a388-77670294a6ca",
    "room_id": "1",
    "user_id": "60610",
    "username": "user60610",
    "message": "Sample message 24 - b2939751-37cc-4fcc-8d5d-0563c7089f19",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.35.89",
    "created_at": "2026-03-29T08:49:36.794Z"
  },
  {
```

```
    },
    {
      "message_id": "c1abd14d-d394-45b1-b414-cae75abedecc",
      "room_id": "1",
      "user_id": "92752",
      "username": "user92752",
      "message": "Sample message 31 - 28f8d5d2-c4f2-486a-99e5-46ffaec3b24e",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.35.89",
      "created_at": "2026-03-29T08:49:36.794Z"
    }
  ]
},
"Q2_userMessageHistory": {
  "description": "Get user's message history",
  "performanceTarget": "< 200 ms",
  "userId": "19763",
  "startTime": "2026-03-29T08:49:36.703735Z",
  "endTime": "2026-03-29T08:55:30.937919Z",
  "messageCount": 3,
  "sampleMessages": [
    {
      "message_id": "13585a17-9ec6-48eb-b1ee-6f03c7374b05",
      "room_id": "16",
      "user_id": "19763",
      "username": "user19763",
      "message": "Sample message 39 - e7995e3f-7295-458f-8f9a-7f1498035cc5",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.30.167",
      "created_at": "2026-03-29T08:55:30.135Z"
    },
    {
      "message_id": "d9143fbd-792d-45e6-83e1-f35245ab2e31",
      "room_id": "4",
      "user_id": "19763",
      "username": "user19763",
      "message": "Sample message 15 - 3d70dcda-c5cf-4b09-8fc0-0fefd6009d22",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.35.89",
      "created_at": "2026-03-29T08:49:37.148Z"
    },
    {
      "message_id": "e50cf54c-c883-444d-8c53-5e6d16997de7",
      "room_id": "6",
      "user_id": "19763",
      "username": "user19763",
      "message": "Sample message 10 - 0f8fba43-3a22-45e0-8901-409b0efb033f",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.35.89",
      "created_at": "2026-03-29T08:49:36.780Z"
    }
  ]
```

```
  },
  "Q3_activeUsersInWindow": {
    "description": "Count active users in time window",
    "performanceTarget": "< 500 ms",
    "startTime": "2026-03-29T08:49:36.703735Z",
    "endTime": "2026-03-29T08:55:30.937919Z",
    "uniqueUserCount": 3238
  },
  "Q4_roomsForUser": {
    "description": "Get rooms user has participated in",
    "performanceTarget": "< 50 ms",
    "userId": "19763",
    "rooms": [
      {
        "room_id": "16",
        "last_activity": "2026-03-29T08:55:30.135Z",
        "message_count": 1
      },
      {
        "room_id": "4",
        "last_activity": "2026-03-29T08:49:37.148Z",
        "message_count": 1
      },
      {
        "room_id": "6",
        "last_activity": "2026-03-29T08:49:36.780Z",
        "message_count": 1
      }
    ]
  }
},
"analyticsQueries": {
  "A1_messagesPerTimeUnit": {
    "description": "Messages per second/minute statistics",
    "perMinute": [
      {
        "bucket": "2026-03-29T08:49:00.000Z",
        "message_count": 1535
      },
      {
        "bucket": "2026-03-29T08:55:00.000Z",
        "message_count": 1762
      }
    ],
    "perSecond": [
      {
        "bucket": "2026-03-29T08:49:36.000Z",
        "message_count": 491
      },
      {
        "bucket": "2026-03-29T08:49:37.000Z",
        "message_count": 1044
      },
      {
        "bucket": "2026-03-29T08:55:29.000Z",
        "message_count": 950
      },
```

```
      {
        "bucket": "2026-03-29T08:55:30.000Z",
        "message_count": 812
      }
    ]
  },
  "A2_topActiveUsers": {
    "description": "Top 10 most active users",
    "topN": 10,
    "users": [
      {
        "user_id": "19763",
        "username": "user19763",
        "message_count": 3,
        "last_active": "2026-03-29T08:55:30.135Z"
      },
      {
        "user_id": "6519",
        "username": "user6519",
        "message_count": 2,
        "last_active": "2026-03-29T08:49:37.076Z"
      },
      {
        "user_id": "80567",
        "username": "user80567",
        "message_count": 2,
        "last_active": "2026-03-29T08:49:37.393Z"
      },
      {
        "user_id": "38325",
        "username": "user38325",
        "message_count": 2,
        "last_active": "2026-03-29T08:55:30.152Z"
      },
      {
        "user_id": "16973",
        "username": "user16973",
        "message_count": 2,
        "last_active": "2026-03-29T08:55:30.226Z"
      },
      {
        "user_id": "72229",
        "username": "user72229",
        "message_count": 2,
        "last_active": "2026-03-29T08:49:37.669Z"
      },
      {
        "user_id": "57142",
        "username": "user57142",
        "message_count": 2,
        "last_active": "2026-03-29T08:55:30.172Z"
      },
      {
        "user_id": "83329",
        "username": "user83329",
        "message_count": 2,
        "last_active": "2026-03-29T08:55:30.158Z"
```

```
      },
      {
        "user_id": "580",
        "username": "user580",
        "message_count": 2,
        "last_active": "2026-03-29T08:55:29.959Z"
      },
      {
        "user_id": "72453",
        "username": "user72453",
        "message_count": 2,
        "last_active": "2026-03-29T08:49:37.142Z"
      }
    ]
  },
  "A3_topActiveRooms": {
    "description": "Top 10 most active rooms",
    "topN": 10,
    "rooms": [
      {
        "room_id": "6",
        "message_count": 274,
        "last_activity": "2026-03-29T08:55:30.856Z"
      },
      {
        "room_id": "18",
        "message_count": 230,
        "last_activity": "2026-03-29T08:55:30.861Z"
      },
      {
        "room_id": "20",
        "message_count": 211,
        "last_activity": "2026-03-29T08:55:30.893Z"
      },
      {
        "room_id": "8",
        "message_count": 197,
        "last_activity": "2026-03-29T08:55:30.858Z"
      },
      {
        "room_id": "2",
        "message_count": 191,
        "last_activity": "2026-03-29T08:55:30.895Z"
      },
      {
        "room_id": "11",
        "message_count": 186,
        "last_activity": "2026-03-29T08:55:30.858Z"
      },
      {
        "room_id": "3",
        "message_count": 180,
        "last_activity": "2026-03-29T08:55:30.937Z"
      },
      {
        "room_id": "9",
        "message_count": 169,
```

```
          "last_activity": "2026-03-29T08:55:30.858Z"
        },
        {
          "room_id": "5",
          "message_count": 164,
          "last_activity": "2026-03-29T08:55:30.856Z"
        },
        {
          "room_id": "15",
          "message_count": 158,
          "last_activity": "2026-03-29T08:55:30.861Z"
        }
      ]
    },
    "A4_userParticipationPatterns": {
      "description": "User participation patterns (rooms visited distribution)",
      "distribution": [
        {
          "rooms_visited": 1,
          "user_count": 3182
        },
        {
          "rooms_visited": 2,
          "user_count": 55
        },
        {
          "rooms_visited": 3,
          "user_count": 1
        }
      ]
    }
  }
}
==== End Metrics API Response ====
```

Part 2 Screenshots:

Test 1: (batchSize=1--, flushintervalMs = 100)


```
==== Metrics API Response ====
{
  "generatedAt": "2026-03-30T00:02:12.421715032Z",
  "viewsLastRefreshed": "2026-03-30T00:01:55.550428661Z",
  "totalMessagesInDB": 1291,
  "queryWindowStart": "2026-03-30T00:02:09.246790Z",
  "queryWindowEnd": "2026-03-30T00:02:11.303665Z",
  "queriedRoomId": "1",
  "queriedUserId": "75816",
  "coreQueries": {
    "Q1_roomMessagesInTimeRange": {
      "description": "Get messages for a room in time range",
      "performanceTarget": "< 100 ms for 1 000 messages",
      "roomId": "1",
      "startTime": "2026-03-30T00:02:09.246790Z",
      "endTime": "2026-03-30T00:02:11.303665Z",
      "messageCount": 79,
      "sampleMessages": [
        {
          "message_id": "0dc78d86-ee8a-453f-9997-f494db5a4cdc",
          "room_id": "1",
          "user_id": "80172",
          "username": "user80172",
          "message": "Sample message 40 - becfdc4d-fbbc-41ce-9d18-06c76b134a38",
          "message_type": "TEXT",
          "server_id": "server-v2",
          "client_ip": "172.31.35.236",
          "created_at": "2026-03-30T00:02:09.344Z"
        },
```

Test 2: (batchSize=500, flushIntervalMs=500)

```
==== Metrics API Response ====
{
  "generatedAt": "2026-03-30T00:48:40.377186860Z",
  "viewsLastRefreshed": "2026-03-30T00:48:13.780444216Z",
  "totalMessagesInDB": 1344,
  "queryWindowStart": "2026-03-30T00:48:37.235299Z",
  "queryWindowEnd": "2026-03-30T00:48:38.542573Z",
  "queriedRoomId": "1",
  "queriedUserId": "81938",
  "coreQueries": {
    "Q1_roomMessagesInTimeRange": {
      "description": "Get messages for a room in time range",
      "performanceTarget": "< 100 ms for 1 000 messages",
      "roomId": "1",
      "startTime": "2026-03-30T00:48:37.235299Z",
      "endTime": "2026-03-30T00:48:38.542573Z",
      "messageCount": 88,
      "sampleMessages": [
        {
          "message_id": "f0d43516-80f8-408c-8266-8a8b192ec03e",
          "room_id": "1",
          "user_id": "32035",
          "username": "user32035",
          "message": "Sample message 7 - 442d47cc-67f8-4168-8281-eca23a1b69b8",
          "message_type": "LEAVE",
          "server_id": "server-v2",
          "client_ip": "172.31.30.167",
          "created_at": "2026-03-30T00:48:37.308Z"
        },
        {
          "message_id": "52f98110-8e81-43b5-a02f-fef812e1a650",
          "room_id": "1",
          "user_id": "85728",
          "username": "user85728",
          "message": "Sample message 30 - 58916747-801c-45d2-b0e2-eb29a9ffa3fd",
          "message_type": "TEXT",
          "server_id": "server-v2",
          "client_ip": "172.31.30.167",
          "created_at": "2026-03-30T00:48:37.309Z"
        },
        {
          "message_id": "febc353f-524c-42af-91ce-f850f4e6c3ae",
          "room_id": "1",
          "user_id": "26424",
          "username": "user26424",
          "message": "Sample message 34 - c31019d3-c0f4-4f7c-8483-1ea68a908978",
          "message_type": "TEXT",
          "server_id": "server-v2",
          "client_ip": "172.31.30.167",
          "created_at": "2026-03-30T00:48:37.310Z"
        },
```

```
{
  "message_id": "ad7fe701-edb7-4c12-92bf-be2a719a34cd",
  "room_id": "1",
  "user_id": "44073",
  "username": "user44073",
  "message": "Sample message 9 - f2df9d7c-e85b-4818-86f9-3405b8f4ef81",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.311Z"
},
{
  "message_id": "b41817d9-de5a-4e56-9b8e-8c803bf1c8da",
  "room_id": "1",
  "user_id": "68142",
  "username": "user68142",
  "message": "Sample message 34 - c31019d3-c0f4-4f7c-8483-1ea68a908978",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.311Z"
},
{
  "message_id": "57378a89-54cc-46fe-a44e-7e3011099d77",
  "room_id": "1",
  "user_id": "54486",
  "username": "user54486",
  "message": "Sample message 23 - a6d52502-6604-46f6-88f0-c9d9dd760317",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.312Z"
},
{
  "message_id": "c600fdff-5ad8-43b3-96b0-5330f6c382f6",
  "room_id": "1",
  "user_id": "61108",
  "username": "user61108",
  "message": "Sample message 33 - 54f9579b-d2ea-4f81-a0d9-335ea919efcd",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.313Z"
},
{
  "message_id": "23d17043-1e5a-4e5b-85a6-1a0e460e1ccd",
  "room_id": "1",
  "user_id": "82960",
  "username": "user82960",
  "message": "Sample message 5 - a0a3b655-dc6e-4644-a9b5-eed4067c708e",
  "message_type": "TEXT",
```

```
    "server_id": "server-v2",
    "client_ip": "172.31.30.167",
    "created_at": "2026-03-30T00:48:37.313Z"
  },
  {
    "message_id": "80a21c25-e2b7-4b55-b41b-a10af43ce2f6",
    "room_id": "1",
    "user_id": "21345",
    "username": "user21345",
    "message": "Sample message 2 - 6b19ff3d-3529-4104-ae95-3a981ba81824",
    "message_type": "LEAVE",
    "server_id": "server-v2",
    "client_ip": "172.31.30.167",
    "created_at": "2026-03-30T00:48:37.314Z"
  },
  {
    "message_id": "f5b0e6fb-dd40-4fcd-bb56-ae5620259447",
    "room_id": "1",
    "user_id": "34810",
    "username": "user34810",
    "message": "JOIN",
    "message_type": "JOIN",
    "server_id": "server-v2",
    "client_ip": "172.31.30.167",
    "created_at": "2026-03-30T00:48:37.406Z"
  },
  {
    "message_id": "41c51786-8828-444e-8189-f849a64125ff",
    "room_id": "1",
    "user_id": "33671",
    "username": "user33671",
    "message": "JOIN",
    "message_type": "JOIN",
    "server_id": "server-v2",
    "client_ip": "172.31.30.167",
    "created_at": "2026-03-30T00:48:37.411Z"
  },
  {
    "message_id": "6f60e14b-a6a4-409f-8763-5b85e82cedcf",
    "room_id": "1",
    "user_id": "24792",
    "username": "user24792",
    "message": "Sample message 41 - 86bce834-41c8-4c29-a3a2-ca71ab713b1f",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.30.167",
    "created_at": "2026-03-30T00:48:37.703Z"
  },
  {
    "message_id": "8cf8e3e0-871c-4627-81c5-50d190ef2f45",
    "room_id": "1",
```

```
  "user_id": "27242",
  "username": "user27242",
  "message": "Sample message 24 - b329b7df-c281-4166-aeca-a6f3decbf09a",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.703Z"
},
{
  "message_id": "6d6d7571-92b0-4364-9e4e-4147ade5fcf3",
  "room_id": "1",
  "user_id": "45045",
  "username": "user45045",
  "message": "Sample message 42 - 7e0f5388-d701-42c1-8751-163f4150defa",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.703Z"
},
{
  "message_id": "76a703cf-f114-4521-acb4-e9ec34e82c37",
  "room_id": "1",
  "user_id": "68606",
  "username": "user68606",
  "message": "Sample message 28 - adaffdda-4aef-4f6f-b332-5ad58812633e",
  "message_type": "LEAVE",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.703Z"
},
{
  "message_id": "0bb83aff-9f68-4364-b2b5-f43fa7fe7cfa",
  "room_id": "1",
  "user_id": "59310",
  "username": "user59310",
  "message": "Sample message 44 - 45db9b31-4944-4744-a9a7-dfdb430d8d1d",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.763Z"
},
{
  "message_id": "4b94fb97-2c13-4cb5-88ae-6f817cbb95e1",
  "room_id": "1",
  "user_id": "84339",
  "username": "user84339",
  "message": "Sample message 4 - d7b5d839-f810-4629-b7e6-3c1bfeb26cc7",
  "message_type": "TEXT",
  "server_id": "server-v2",
  "client_ip": "172.31.30.167",
  "created_at": "2026-03-30T00:48:37.763Z"
```

```
    },
    {
      "message_id": "5c74dbd0-c326-42fb-b52d-4e47bc7d60cd",
      "room_id": "1",
      "user_id": "31510",
      "username": "user31510",
      "message": "Sample message 24 - b329b7df-c281-4166-aeca-a6f3decbf09a",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.30.167",
      "created_at": "2026-03-30T00:48:37.763Z"
    },
    {
      "message_id": "e8c30f75-0c98-45fc-bb14-54351ccaafcf",
      "room_id": "1",
      "user_id": "50075",
      "username": "user50075",
      "message": "Sample message 50 - 36c83bdb-d4b5-4d3f-bb42-aad0e7f780a3",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.30.167",
      "created_at": "2026-03-30T00:48:37.763Z"
    },
    {
      "message_id": "3e40b3e6-a711-4506-bc29-d1562c451874",
      "room_id": "1",
      "user_id": "47309",
      "username": "user47309",
      "message": "Sample message 30 - 58916747-801c-45d2-b0e2-eb29a9ffa3fd",
      "message_type": "TEXT",
      "server_id": "server-v2",
      "client_ip": "172.31.30.167",
      "created_at": "2026-03-30T00:48:37.763Z"
    }
  ]
},
"Q2_userMessageHistory": {
  "description": "Get user's message history",
  "performanceTarget": "< 200 ms",
  "userId": "81938",
  "startTime": "2026-03-30T00:48:37.235299Z",
  "endTime": "2026-03-30T00:48:38.542573Z",
  "messageCount": 2,
  "sampleMessages": [
    {
      "message_id": "52ff0ee9-da72-4195-a457-7226e5b7d668",
      "room_id": "17",
      "user_id": "81938",
      "username": "user81938",
      "message": "Sample message 1 - d1709e44-32cd-4d0d-b7f5-843d56133fca",
      "message_type": "TEXT",
```

```
        "server_id": "server-v2",
        "client_ip": "172.31.30.167",
        "created_at": "2026-03-30T00:48:37.737Z"
      },
      {
        "message_id": "78106a35-c8af-4d52-bb5d-cc08a26b89af",
        "room_id": "4",
        "user_id": "81938",
        "username": "user81938",
        "message": "Sample message 30 - 58916747-801c-45d2-b0e2-eb29a9ffa3fd",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.30.167",
        "created_at": "2026-03-30T00:48:37.319Z"
      }
    ]
  },
  "Q3_activeUsersInWindow": {
    "description": "Count active users in time window",
    "performanceTarget": "< 500 ms",
    "startTime": "2026-03-30T00:48:37.235299Z",
    "endTime": "2026-03-30T00:48:38.542573Z",
    "uniqueUserCount": 1338
  },
  "Q4_roomsForUser": {
    "description": "Get rooms user has participated in",
    "performanceTarget": "< 50 ms",
    "userId": "81938",
    "rooms": [
      {
        "room_id": "17",
        "last_activity": "2026-03-30T00:48:37.737Z",
        "message_count": 1
      },
      {
        "room_id": "4",
        "last_activity": "2026-03-30T00:48:37.319Z",
        "message_count": 1
      }
    ]
  }
},
"analyticsQueries": {
  "A1_messagesPerTimeUnit": {
    "description": "Messages per second/minute statistics",
    "perMinute": [

    ],
    "perSecond": [

    ]
```

```
    },
    "A2_topActiveUsers": {
      "description": "Top 10 most active users",
      "topN": 10,
      "users": [

      ]
    },
    "A3_topActiveRooms": {
      "description": "Top 10 most active rooms",
      "topN": 10,
      "rooms": [

      ]
    },
    "A4_userParticipationPatterns": {
      "description": "User participation patterns (rooms visited distribution)",
      "distribution": [

      ]
    }
  }
}
==== End Metrics API Response ====
```


T3 Testing (batchSize: 1000, flushIntervalMs: 500, Messages published: 50,000):

```
Every 2.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c 'SELECT COUNT(*) FROM messages;' ip-172-31-59-20: Mon Mar 30 02:48:17 2026

 count
-------
 51114
(1 row)
```

```
ubuntu@ip-172-31-59-20:~$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "generatedAt": "2026-03-30T02:49:06.431146166Z",
    "viewsLastRefreshed": "2026-03-30T02:48:45.105754556Z",
    "totalMessagesInDB": 51114,
    "queryWindowStart": "2026-03-30T01:50:45.487323Z",
    "queryWindowEnd": "2026-03-30T02:42:39.991513Z",
    "queriedRoomId": "1",
    "queriedUserId": "9389",
    "coreQueries": {
        "Q1_roomMessagesInTimeRange": {
            "description": "Get messages for a room in time range",
            "performanceTarget": "< 100 ms for 1 000 messages",
            "roomId": "1",
            "startTime": "2026-03-30T01:50:45.487323Z",
            "endTime": "2026-03-30T02:42:39.991513Z",
            "messageCount": 2657,
            "sampleMessages": [
                {
                    "message_id": "c9f975b9-1622-43f9-81ff-eefecf260f65",
                    "room_id": "1",
                    "user_id": "69135",
                    "username": "user69135",
                    "message": "JOIN",
                    "message_type": "JOIN",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T01:50:45.607Z"
                },
                {
                    "message_id": "eeb93fa3-9060-4071-85b9-4313687680ac",
                    "room_id": "1",
                    "user_id": "83426",
                    "username": "user83426",
                    "message": "Sample message 44 - 9f984b89-765c-4bf0-b86b-66b65fdfadf5",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T01:50:45.607Z"
                },
                {
                    "message_id": "b3129bde-3bed-4f37-b00b-0c978c6577a0",
                    "room_id": "1",
                    "user_id": "75139",
                    "username": "user75139",
                    "message": "Sample message 30 - c854f801-6858-49d5-9b78-0b4470d42352",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T01:50:45.607Z"
                },
                {
                    "message_id": "2ae96dcb-7f61-4cf0-8e77-0601bd2daff3",
                    "room_id": "1",
                    "user_id": "7061",
                    "username": "user7061",
```

```
        "message": "Sample message 50 - df1fde1f-4eac-49c0-9138-e382d3e189d8",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.608Z"
    },
    {
        "message_id": "435ffdfc-d679-4eb9-9ac2-4a57f24431da",
        "room_id": "1",
        "user_id": "59746",
        "username": "user59746",
        "message": "JOIN",
        "message_type": "JOIN",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.608Z"
    },
    {
        "message_id": "91fbd776-bbc4-4215-967c-ade416e0de56",
        "room_id": "1",
        "user_id": "70434",
        "username": "user70434",
        "message": "Sample message 12 - 9d49e7a9-c96b-49e7-90c4-a05dfdc9611e",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "93253d42-fb7b-4eb9-95ec-292a741f96ca",
        "room_id": "1",
        "user_id": "32044",
        "username": "user32044",
        "message": "Sample message 37 - 10ec5756-a002-49f8-8475-4491434b77a6",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "bd36639f-2ece-4009-91d1-5833e0979880",
        "room_id": "1",
        "user_id": "37619",
        "username": "user37619",
        "message": "Sample message 17 - 8ee64e25-9a77-4bcf-85dc-bd6b6d67d8c6",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "9747f147-6e60-4394-8bc6-7f9a4fac9328",
        "room_id": "1",
        "user_id": "90062",
        "username": "user90062",
        "message": "Sample message 21 - a6a7cc9e-1021-4ec0-b8bb-098a0ae5344e",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "be8ce686-26d0-4187-9b88-7bb0698cd03b",
        "room_id": "1",
        "user_id": "91082",
```

```
        "username": "user91082",
        "message": "Sample message 37 - 10ec5756-a002-49f8-8475-4491434b77a6",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "831bdaf7-1c3e-4c15-be78-4968e42b4628",
        "room_id": "1",
        "user_id": "58702",
        "username": "user58702",
        "message": "Sample message 28 - 852c8fa6-e6db-48a8-b4d8-ee3147606bfb",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.609Z"
    },
    {
        "message_id": "e6169f76-9ef7-45ed-b8dd-20e33370a5d3",
        "room_id": "1",
        "user_id": "87214",
        "username": "user87214",
        "message": "Sample message 43 - 591efc0f-1696-412d-9f19-dbeca1fa2fc9",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.610Z"
    },
    {
        "message_id": "b052df1d-017d-4bae-9dec-d4c98181eba3",
        "room_id": "1",
        "user_id": "94239",
        "username": "user94239",
        "message": "Sample message 33 - b4d585b9-5ff1-453a-bf5c-33c95ff2ed8c",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.610Z"
    },
    {
        "message_id": "b2a7526a-f5f3-4ae9-b951-bb63e754559e",
        "room_id": "1",
        "user_id": "14478",
        "username": "user14478",
        "message": "Sample message 33 - b4d585b9-5ff1-453a-bf5c-33c95ff2ed8c",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.610Z"
    },
    {
        "message_id": "6726714b-41ee-4e39-9548-88fdbc09bede",
        "room_id": "1",
        "user_id": "57608",
        "username": "user57608",
        "message": "Sample message 6 - f16d7f1a-73b5-4306-af74-3e829e9ce0fc",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T01:50:45.610Z"
    },
    {
```

```
        {
            "message_id": "d2234e7d-e756-4a9d-bc9b-6422e90a1add",
            "room_id": "1",
            "user_id": "89323",
            "username": "user89323",
            "message": "Sample message 23 - 4fef1880-f9e9-43d3-b4c4-94d9d4469938",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T01:50:45.610Z"
        },
        {
            "message_id": "75613475-a8bc-4cb5-9f8d-96e9baafb01f",
            "room_id": "1",
            "user_id": "40017",
            "username": "user40017",
            "message": "Sample message 29 - 34583de8-0fe5-459b-bb64-f72841f429a0",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T01:50:45.610Z"
        },
        {
            "message_id": "17e19de4-6192-4544-9121-33076d58566a",
            "room_id": "1",
            "user_id": "85166",
            "username": "user85166",
            "message": "Sample message 19 - ddb42604-abbc-404f-b382-abaed15ea808",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T01:50:45.610Z"
        },
        {
            "message_id": "b45a7233-5869-44a0-b668-9cb9fcce5b96",
            "room_id": "1",
            "user_id": "67630",
            "username": "user67630",
            "message": "Sample message 30 - c854f801-6858-49d5-9b78-0b4470d42352",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T01:50:45.610Z"
        },
        {
            "message_id": "e25a68f9-ae4b-4618-b37f-1e4bb36233fc",
            "room_id": "1",
            "user_id": "97851",
            "username": "user97851",
            "message": "Sample message 34 - d75db523-81c9-4c4a-84c5-c274a7620eb6",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T01:50:45.610Z"
        }
    ]
},
"Q2_userMessageHistory": {
    "description": "Get user's message history",
    "performanceTarget": "< 200 ms",
    "userId": "9389",
```

```
"startTime": "2026-03-30T01:50:45.487323Z",
"endTime": "2026-03-30T02:42:39.991513Z",
"messageCount": 6,
"sampleMessages": [
    {
        "message_id": "059d882e-abff-4681-b0fd-2a73afd2780e",
        "room_id": "9",
        "user_id": "9389",
        "username": "user9389",
        "message": "Test message 49388",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:42:39.854Z"
    },
    {
        "message_id": "d09ccae6-88bb-4e20-a597-b686c2f40821",
        "room_id": "9",
        "user_id": "9389",
        "username": "user9389",
        "message": "Test message 39388",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:42:37.286Z"
    },
    {
        "message_id": "31c647f8-0830-44c5-a0fc-8f65b9e3cfc6",
        "room_id": "9",
        "user_id": "9389",
        "username": "user9389",
        "message": "Test message 29388",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:42:33.822Z"
    },
    {
        "message_id": "ae2dc251-5449-4761-986b-718051bc0aab",
        "room_id": "9",
        "user_id": "9389",
        "username": "user9389",
        "message": "Test message 19388",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:42:30.282Z"
    },
    {
        "message_id": "cbc15cb4-b987-40d6-88d5-65baca83b043",
        "room_id": "9",
        "user_id": "9389",
        "username": "user9389",
        "message": "Test message 9388",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:42:26.557Z"
    },
    {
        "message_id": "76304973-a422-419a-8a3d-d4e19dbf2b4c",
        "room_id": "10",
        "user_id": "9389",
```

```
                "username": "user9389",
                "message": "Sample message 13 - 086bd444-6372-48dd-95fc-b43705189b59",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T01:50:45.608Z"
            }
        ]
    },
    "Q3_activeUsersInWindow": {
        "description": "Count active users in time window",
        "performanceTarget": "< 500 ms",
        "startTime": "2026-03-30T01:50:45.487323Z",
        "endTime": "2026-03-30T02:42:39.991513Z",
        "uniqueUserCount": 10991
    },
    "Q4_roomsForUser": {
        "description": "Get rooms user has participated in",
        "performanceTarget": "< 50 ms",
        "userId": "9389",
        "rooms": [
            {
                "room_id": "9",
                "last_activity": "2026-03-30T02:42:39.854Z",
                "message_count": 5
            },
            {
                "room_id": "10",
                "last_activity": "2026-03-30T01:50:45.608Z",
                "message_count": 1
            }
        ]
    }
},
"analyticsQueries": {
    "A1_messagesPerTimeUnit": {
        "description": "Messages per second/minute statistics",
        "perMinute": [
            {
                "bucket": "2026-03-30T01:50:00.000Z",
                "message_count": 1114
            },
            {
                "bucket": "2026-03-30T02:42:00.000Z",
                "message_count": 50000
            }
        ],
        "perSecond": [
            {
                "bucket": "2026-03-30T01:50:45.000Z",
                "message_count": 479
            },
            {
                "bucket": "2026-03-30T01:50:46.000Z",
                "message_count": 174
            },
            {
                "bucket": "2026-03-30T01:50:47.000Z",
                "message_count": 411
            },
            {
                "bucket": "2026-03-30T01:50:48.000Z",
                "message_count": 50
```

```
{
    "bucket": "2026-03-30T02:42:22.000Z",
    "message_count": 794
},
{
    "bucket": "2026-03-30T02:42:23.000Z",
    "message_count": 2157
},
{
    "bucket": "2026-03-30T02:42:24.000Z",
    "message_count": 2319
},
{
    "bucket": "2026-03-30T02:42:25.000Z",
    "message_count": 2697
},
{
    "bucket": "2026-03-30T02:42:26.000Z",
    "message_count": 2601
},
{
    "bucket": "2026-03-30T02:42:27.000Z",
    "message_count": 2703
},
{
    "bucket": "2026-03-30T02:42:28.000Z",
    "message_count": 2595
},
{
    "bucket": "2026-03-30T02:42:29.000Z",
    "message_count": 2766
},
{
    "bucket": "2026-03-30T02:42:30.000Z",
    "message_count": 2852
},
{
    "bucket": "2026-03-30T02:42:31.000Z",
    "message_count": 2873
},
{
    "bucket": "2026-03-30T02:42:32.000Z",
    "message_count": 2647
},
{
    "bucket": "2026-03-30T02:42:33.000Z",
    "message_count": 2992
},
{
    "bucket": "2026-03-30T02:42:34.000Z",
    "message_count": 2832
},
{
    "bucket": "2026-03-30T02:42:35.000Z",
    "message_count": 2592
},
{
    "bucket": "2026-03-30T02:42:36.000Z",
    "message_count": 2894
},
{
    "bucket": "2026-03-30T02:42:37.000Z",
    "message_count": 3615
},
{
```

```
            },
            {
                "bucket": "2026-03-30T02:42:38.000Z",
                "message_count": 4362
            },
            {
                "bucket": "2026-03-30T02:42:39.000Z",
                "message_count": 3709
            }
        ]
    },
    "A2_topActiveUsers": {
        "description": "Top 10 most active users",
        "topN": 10,
        "users": [
            {
                "user_id": "8751",
                "username": "user8751",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:39.721Z"
            },
            {
                "user_id": "2794",
                "username": "user2794",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.199Z"
            },
            {
                "user_id": "6456",
                "username": "user6456",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:39.042Z"
            },
            {
                "user_id": "4447",
                "username": "user4447",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.602Z"
            },
            {
                "user_id": "9530",
                "username": "user9530",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:39.884Z"
            },
            {
                "user_id": "3285",
                "username": "user3285",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.344Z"
            },
            {
                "user_id": "7518",
                "username": "user7518",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:39.319Z"
            },
            {
                "user_id": "4430",
                "username": "user4430",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.598Z"
            },
            {
                "user_id": "2962",
                "username": "user2962",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.248Z"
            },
            {
```

```
            {
                "user_id": "5033",
                "username": "user5033",
                "message_count": 6,
                "last_active": "2026-03-30T02:42:38.732Z"
            }
        ]
    },
    "A3_topActiveRooms": {
        "description": "Top 10 most active rooms",
        "topN": 10,
        "rooms": [
            {
                "room_id": "1",
                "message_count": 2657,
                "last_activity": "2026-03-30T02:42:39.987Z"
            },
            {
                "room_id": "3",
                "message_count": 2624,
                "last_activity": "2026-03-30T02:42:39.988Z"
            },
            {
                "room_id": "20",
                "message_count": 2603,
                "last_activity": "2026-03-30T02:42:39.991Z"
            },
            {
                "room_id": "7",
                "message_count": 2577,
                "last_activity": "2026-03-30T02:42:39.988Z"
            },
            {
                "room_id": "12",
                "message_count": 2566,
                "last_activity": "2026-03-30T02:42:39.989Z"
            },
            {
                "room_id": "8",
                "message_count": 2566,
                "last_activity": "2026-03-30T02:42:39.989Z"
            },
            {
                "room_id": "2",
                "message_count": 2564,
                "last_activity": "2026-03-30T02:42:39.987Z"
            },
            {
                "room_id": "9",
                "message_count": 2563,
                "last_activity": "2026-03-30T02:42:39.989Z"
            },
            {
                "room_id": "4",
                "message_count": 2556,
                "last_activity": "2026-03-30T02:42:39.988Z"
            },
            {
                "room_id": "13",
                "message_count": 2549,
                "last_activity": "2026-03-30T02:42:39.990Z"
            }
        ]
    },
    "A4_userParticipationPatterns": {
        "description": "User participation patterns (rooms visited distribution)",
        "distribution": [
            {
                "rooms_visited": 1,
                "user_count": 10871
```

```
                },
                {
                    "rooms_visited": 2,
                    "user_count": 120
                }
            ]
        }
    }
}
```


T4 screenshots(**batchSize=5000, flushIntervalMs=1000**):

```
Every 2.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c 'SELECT COUNT(*) FROM messages;' ip-172-31-59-20: Mon Mar 30 03:08:55 2026

 count
-------
 50000
(1 row)
```

```
ubuntu@ip-172-31-59-20:~$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "generatedAt": "2026-03-30T03:10:08.473837780Z",
    "viewsLastRefreshed": "2026-03-30T03:09:52.448663232Z",
    "totalMessagesInDB": 50000,
    "queryWindowStart": "2026-03-30T02:55:22.533636Z",
    "queryWindowEnd": "2026-03-30T02:55:41.142542Z",
    "queriedRoomId": "1",
    "queriedUserId": "2571",
    "coreQueries": {
        "Q1_roomMessagesInTimeRange": {
            "description": "Get messages for a room in time range",
            "performanceTarget": "< 100 ms for 1 000 messages",
            "roomId": "1",
            "startTime": "2026-03-30T02:55:22.533636Z",
            "endTime": "2026-03-30T02:55:41.142542Z",
            "messageCount": 2500,
            "sampleMessages": [
                {
                    "message_id": "485279b1-2c6b-41d8-8bdb-d5772be6d151",
                    "room_id": "1",
                    "user_id": "1",
                    "username": "user1",
                    "message": "Test message 0",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T02:55:22.533Z"
                },
                {
                    "message_id": "19ecb010-eeb6-410c-936d-d63157480459",
                    "room_id": "1",
                    "user_id": "21",
                    "username": "user21",
                    "message": "Test message 20",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T02:55:22.537Z"
                },
                {
                    "message_id": "d833035f-133b-4451-8397-16e071ff2a66",
                    "room_id": "1",
                    "user_id": "41",
                    "username": "user41",
                    "message": "Test message 40",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T02:55:22.547Z"
                },
```

```
    {
        "message_id": "0db19643-6672-4ff6-9542-a5382f2e9a03",
        "room_id": "1",
        "user_id": "61",
        "username": "user61",
        "message": "Test message 60",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.563Z"
    },
    {
        "message_id": "6759bd07-fea4-4859-953a-ce65da49e2c0",
        "room_id": "1",
        "user_id": "81",
        "username": "user81",
        "message": "Test message 80",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.572Z"
    },
    {
        "message_id": "2fad6f99-a63a-46bb-9fcb-78466578c912",
        "room_id": "1",
        "user_id": "101",
        "username": "user101",
        "message": "Test message 100",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.576Z"
    },
    {
        "message_id": "4b5a9db7-5af2-4ec0-b7f7-86f98c938e27",
        "room_id": "1",
        "user_id": "121",
        "username": "user121",
        "message": "Test message 120",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.592Z"
    },
    {
        "message_id": "3680589e-5212-42d1-b465-3947fcee9a6a",
        "room_id": "1",
        "user_id": "141",
        "username": "user141",
        "message": "Test message 140",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.604Z"
    },
    {
        "message_id": "8f1d4a1e-c64e-4ba7-b21e-524fa4bf6477",
        "room_id": "1",
        "user_id": "161",
        "username": "user161",
        "message": "Test message 160",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.610Z"
    },
    {
        "message_id": "ea840462-784a-4b08-ba3d-76ed38073026",
        "room_id": "1",
        "user_id": "181",
        "username": "user181",
        "message": "Test message 180",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.616Z"
    },
    {
        "message_id": "483091d8-8461-4803-a299-bbd28393caef",
        "room_id": "1",
        "user_id": "201",
        "username": "user201",
```

```
        "message": "Test message 200",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.624Z"
    },
    {
        "message_id": "a58cf642-abfb-48ad-bbe9-026d13e27275",
        "room_id": "1",
        "user_id": "221",
        "username": "user221",
        "message": "Test message 220",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.631Z"
    },
    {
        "message_id": "13de11b8-f5f0-4ced-86bb-f1fa276b4cbc",
        "room_id": "1",
        "user_id": "241",
        "username": "user241",
        "message": "Test message 240",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.638Z"
    },
    {
        "message_id": "27ff946e-2c06-423d-8f8b-f24855cae51c",
        "room_id": "1",
        "user_id": "261",
        "username": "user261",
        "message": "Test message 260",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.642Z"
    },
    {
        "message_id": "b65561bf-47c4-4d23-8149-2d69a7f1a89a",
        "room_id": "1",
        "user_id": "281",
        "username": "user281",
        "message": "Test message 280",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.646Z"
    },
    {
        "message_id": "8115e352-04f9-4702-935d-91cab373c743",
        "room_id": "1",
        "user_id": "301",
        "username": "user301",
        "message": "Test message 300",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.656Z"
    },
    {
        "message_id": "fb4509cd-bf86-4fd2-bf49-5c1dc44b0379",
        "room_id": "1",
        "user_id": "321",
        "username": "user321",
        "message": "Test message 320",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T02:55:22.667Z"
    },
    {
        "message_id": "afb720f0-7be0-4ed8-b8ec-002f210c1598",
        "room_id": "1",
        "user_id": "341",
        "username": "user341",
        "message": "Test message 340",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
```

```
                "created_at": "2026-03-30T02:55:22.680Z"
            },
            {
                "message_id": "890c71e7-b5a4-43b4-812a-f0e0aeba3f97",
                "room_id": "1",
                "user_id": "361",
                "username": "user361",
                "message": "Test message 360",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:22.696Z"
            },
            {
                "message_id": "056000cb-e70d-4022-b62e-f051ee41aa6a",
                "room_id": "1",
                "user_id": "381",
                "username": "user381",
                "message": "Test message 380",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:22.709Z"
            }
        ]
    },
    "Q2_userMessageHistory": {
        "description": "Get user's message history",
        "performanceTarget": "< 200 ms",
        "userId": "2571",
        "startTime": "2026-03-30T02:55:22.533636Z",
        "endTime": "2026-03-30T02:55:41.142542Z",
        "messageCount": 5,
        "sampleMessages": [
            {
                "message_id": "1bb37188-3e06-4680-8287-c953890460a5",
                "room_id": "11",
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 42570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:38.506Z"
            },
            {
                "message_id": "28eaed27-d35e-4b06-8a09-583dab968b4c",
                "room_id": "11",
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 32570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:35.021Z"
            },
            {
                "message_id": "e4a28b1c-58d8-49c8-b598-e2513d1ba833",
                "room_id": "11",
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 22570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:31.633Z"
            },
            {
                "message_id": "6717f37a-636a-4cac-a6e5-a4113356d024",
                "room_id": "11",
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 12570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:28.187Z"
            },
            {
                "message_id": "9ee5215e-9a47-4b7d-bd84-b985e719783e",
                "room_id": "11",
```

```
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 2570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T02:55:24.226Z"
            }
        ]
    },
    "Q3_activeUsersInWindow": {
        "description": "Count active users in time window",
        "performanceTarget": "< 500 ms",
        "startTime": "2026-03-30T02:55:22.533636Z",
        "endTime": "2026-03-30T02:55:41.142542Z",
        "uniqueUserCount": 10000
    },
    "Q4_roomsForUser": {
        "description": "Get rooms user has participated in",
        "performanceTarget": "< 50 ms",
        "userId": "2571",
        "rooms": [
            {
                "room_id": "11",
                "last_activity": "2026-03-30T02:55:38.506Z",
                "message_count": 5
            }
        ]
    }
},
"analyticsQueries": {
    "A1_messagesPerTimeUnit": {
        "description": "Messages per second/minute statistics",
        "perMinute": [
            {
                "bucket": "2026-03-30T02:55:00.000Z",
                "message_count": 50000
            }
        ],
        "perSecond": [
            {
                "bucket": "2026-03-30T02:55:22.000Z",
                "message_count": 716
            },
            {
                "bucket": "2026-03-30T02:55:23.000Z",
                "message_count": 1476
            },
            {
                "bucket": "2026-03-30T02:55:24.000Z",
                "message_count": 2074
            },
            {
                "bucket": "2026-03-30T02:55:25.000Z",
                "message_count": 2349
            },
            {
                "bucket": "2026-03-30T02:55:26.000Z",
                "message_count": 2641
            },
            {
                "bucket": "2026-03-30T02:55:27.000Z",
                "message_count": 2826
            },
            {
                "bucket": "2026-03-30T02:55:28.000Z",
                "message_count": 2854
            },
            {
                "bucket": "2026-03-30T02:55:29.000Z",
                "message_count": 2960
            },
            {
                "bucket": "2026-03-30T02:55:30.000Z",
                "message_count": 2880
            },
            {
                "bucket": "2026-03-30T02:55:31.000Z",
                "message_count": 2916
            },
            {
```

```
            {
                "bucket": "2026-03-30T02:55:32.000Z",
                "message_count": 3023
            },
            {
                "bucket": "2026-03-30T02:55:33.000Z",
                "message_count": 2959
            },
            {
                "bucket": "2026-03-30T02:55:34.000Z",
                "message_count": 2843
            },
            {
                "bucket": "2026-03-30T02:55:35.000Z",
                "message_count": 2864
            },
            {
                "bucket": "2026-03-30T02:55:36.000Z",
                "message_count": 2828
            },
            {
                "bucket": "2026-03-30T02:55:37.000Z",
                "message_count": 2912
            },
            {
                "bucket": "2026-03-30T02:55:38.000Z",
                "message_count": 2945
            },
            {
                "bucket": "2026-03-30T02:55:39.000Z",
                "message_count": 3011
            },
            {
                "bucket": "2026-03-30T02:55:40.000Z",
                "message_count": 2608
            },
            {
                "bucket": "2026-03-30T02:55:41.000Z",
                "message_count": 315
            }
        ]
    },
    "A2_topActiveUsers": {
        "description": "Top 10 most active users",
        "topN": 10,
        "users": [
            {
                "user_id": "3542",
                "username": "user3542",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.813Z"
            },
            {
                "user_id": "3226",
                "username": "user3226",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.716Z"
            },
            {
                "user_id": "247",
                "username": "user247",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:37.688Z"
            },
            {
                "user_id": "2275",
                "username": "user2275",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.405Z"
            },
            {
                "user_id": "3202",
                "username": "user3202",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.709Z"
            },
            {
                "user_id": "8362",
                "username": "user8362",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:40.484Z"
            },
```

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            },
            {
                "user_id": "1822",
                "username": "user1822",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.233Z"
            },
            {
                "user_id": "3781",
                "username": "user3781",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.902Z"
            },
            {
                "user_id": "8902",
                "username": "user8902",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:40.640Z"
            },
            {
                "user_id": "2571",
                "username": "user2571",
                "message_count": 5,
                "last_active": "2026-03-30T02:55:38.506Z"
            }
        ]
    },
    "A3_topActiveRooms": {
        "description": "Top 10 most active rooms",
        "topN": 10,
        "rooms": [
            {
                "room_id": "1",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.133Z"
            },
            {
                "room_id": "4",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.136Z"
            },
            {
                "room_id": "9",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.138Z"
            },
            {
                "room_id": "8",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.138Z"
            },
            {
                "room_id": "12",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.139Z"
            },
            {
                "room_id": "10",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.139Z"
            },
            {
                "room_id": "7",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.137Z"
            },
            {
                "room_id": "13",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.141Z"
            },
            {
                "room_id": "17",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.142Z"
            },
            {
                "room_id": "15",
                "message_count": 2500,
                "last_activity": "2026-03-30T02:55:41.141Z"
            }
        ]
```

```
        },
        "A4_userParticipationPatterns": {
            "description": "User participation patterns (rooms visited distribution)",
            "distribution": [
                {
                    "rooms_visited": 1,
                    "user_count": 10000
                }
            ]
        }
    }
}
ubuntu@ip-172-31-59-20:~$
```

T5 screenshots(batchSize =1000, flushIntervalMs = 100):

```
Every 2.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c 'SELECT COUNT(*) FROM messages;'        ip-172-31-59-20: Mon Mar 30 03:18:32 2026

 count
-------
 50000
(1 row)
```

```
ubuntu@ip-172-31-59-20:~$ curl -s http://localhost:8082/api/metrics | python3 -m json.tool
{
    "generatedAt": "2026-03-30T03:49:25.207812056Z",
    "viewsLastRefreshed": "2026-03-30T03:49:13.354990351Z",
    "totalMessagesInDB": 50000,
    "queryWindowStart": "2026-03-30T03:16:55.693839Z",
    "queryWindowEnd": "2026-03-30T03:17:15.779312Z",
    "queriedRoomId": "1",
    "queriedUserId": "2571",
    "coreQueries": {
        "Q1_roomMessagesInTimeRange": {
            "description": "Get messages for a room in time range",
            "performanceTarget": "< 100 ms for 1 000 messages",
            "roomId": "1",
            "startTime": "2026-03-30T03:16:55.693839Z",
            "endTime": "2026-03-30T03:17:15.779312Z",
            "messageCount": 2500,
            "sampleMessages": [
                {
                    "message_id": "eeac7311-4690-4bb3-bf1c-725057b9397f",
                    "room_id": "1",
                    "user_id": "1",
                    "username": "user1",
                    "message": "Test message 0",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T03:16:55.693Z"
                },
                {
                    "message_id": "80c72610-f03a-4bb5-b479-41e92443c560",
                    "room_id": "1",
                    "user_id": "21",
                    "username": "user21",
                    "message": "Test message 20",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T03:16:55.703Z"
                },
                {
                    "message_id": "c42e9a31-69a1-4cab-8e8f-4dc0e917d0b6",
                    "room_id": "1",
                    "user_id": "41",
                    "username": "user41",
                    "message": "Test message 40",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T03:16:55.718Z"
                },
                {
                    "message_id": "8e96ccc8-3654-4c3d-982a-12d007842bc6",
                    "room_id": "1",
                    "user_id": "61",
                    "username": "user61",
                    "message": "Test message 60",
                    "message_type": "TEXT",
                    "server_id": "server-v2",
                    "client_ip": "172.31.59.20",
                    "created_at": "2026-03-30T03:16:55.736Z"
                },
                {
```

```
    "message_id": "24bc0b69-347c-4877-84bf-53bccb51e254",
    "room_id": "1",
    "user_id": "81",
    "username": "user81",
    "message": "Test message 80",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
    "created_at": "2026-03-30T03:16:55.755Z"
},
{
    "message_id": "d0e48cd2-8640-4501-8097-ce862cf4cb93",
    "room_id": "1",
    "user_id": "101",
    "username": "user101",
    "message": "Test message 100",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
    "created_at": "2026-03-30T03:16:55.766Z"
},
{
    "message_id": "2c339ac4-75bd-4054-9c2c-f3667a600d25",
    "room_id": "1",
    "user_id": "121",
    "username": "user121",
    "message": "Test message 120",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
    "created_at": "2026-03-30T03:16:55.773Z"
},
{
    "message_id": "1679e9b3-f59f-4dae-9369-af80aa8252d2",
    "room_id": "1",
    "user_id": "141",
    "username": "user141",
    "message": "Test message 140",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
    "created_at": "2026-03-30T03:16:55.780Z"
},
{
    "message_id": "7979cf52-8e94-44b6-897f-2e48ca1fb2bd",
    "room_id": "1",
    "user_id": "161",
    "username": "user161",
    "message": "Test message 160",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
    "created_at": "2026-03-30T03:16:55.794Z"
},
{
    "message_id": "6637d7ee-298f-4430-bf30-c63ab7c42fd5",
    "room_id": "1",
    "user_id": "181",
    "username": "user181",
    "message": "Test message 180",
    "message_type": "TEXT",
    "server_id": "server-v2",
    "client_ip": "172.31.59.20",
```

```
        "created_at": "2026-03-30T03:16:55.815Z"
    },
    {
        "message_id": "5e7d8fdf-efe8-4865-a1ce-9e80257c0813",
        "room_id": "1",
        "user_id": "201",
        "username": "user201",
        "message": "Test message 200",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.828Z"
    },
    {
        "message_id": "1ac0ace9-1dfc-4a89-a80e-5ee2a3c0ea3c",
        "room_id": "1",
        "user_id": "221",
        "username": "user221",
        "message": "Test message 220",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.852Z"
    },
    {
        "message_id": "3014f56d-e6c3-45a8-8722-e0e2ed8bc7d8",
        "room_id": "1",
        "user_id": "241",
        "username": "user241",
        "message": "Test message 240",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.863Z"
    },
    {
        "message_id": "6ce2000f-c132-47dd-9685-858aec79dd8d",
        "room_id": "1",
        "user_id": "261",
        "username": "user261",
        "message": "Test message 260",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.869Z"
    },
    {
        "message_id": "42fcb859-b16b-4d24-a0ea-11f66473ace7",
        "room_id": "1",
        "user_id": "281",
        "username": "user281",
        "message": "Test message 280",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.889Z"
    },
    {
        "message_id": "09bce1d6-77ec-413c-b3e1-0f86064492c3",
        "room_id": "1",
        "user_id": "301",
        "username": "user301",
        "message": "Test message 300",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.911Z"
    },
    {
        "message_id": "b9d372f0-cecd-440a-9b23-0beb3e845026",
        "room_id": "1",
        "user_id": "321",
        "username": "user321",
        "message": "Test message 320",
        "message_type": "TEXT",
        "server_id": "server-v2",
        "client_ip": "172.31.59.20",
        "created_at": "2026-03-30T03:16:55.925Z"
    },
    {
```

```
            "message_id": "bae7363a-3fa0-4aad-b8f3-6d43d592b6da",
            "room_id": "1",
            "user_id": "341",
            "username": "user341",
            "message": "Test message 340",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:16:55.946Z"
        },
        {
            "message_id": "5edce2de-eabb-4a01-b240-d2cbb6020c2b",
            "room_id": "1",
            "user_id": "361",
            "username": "user361",
            "message": "Test message 360",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:16:55.972Z"
        },
        {
            "message_id": "9e4bda1a-3e3f-4200-9504-c801c98ff64c",
            "room_id": "1",
            "user_id": "381",
            "username": "user381",
            "message": "Test message 380",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:16:55.982Z"
        }
    ]
},
"Q2_userMessageHistory": {
    "description": "Get user's message history",
    "performanceTarget": "< 200 ms",
    "userId": "2571",
    "startTime": "2026-03-30T03:16:55.693839Z",
    "endTime": "2026-03-30T03:17:15.779312Z",
    "messageCount": 5,
    "sampleMessages": [
        {
            "message_id": "cfe5082e-232f-46eb-b455-3a3b79e68ea5",
            "room_id": "11",
            "user_id": "2571",
            "username": "user2571",
            "message": "Test message 42570",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:17:13.128Z"
        },
        {
            "message_id": "0e9a5910-18f5-4e1f-9ad1-ed92e649b619",
            "room_id": "11",
            "user_id": "2571",
            "username": "user2571",
            "message": "Test message 32570",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:17:09.390Z"
        },
        {
            "message_id": "16978f89-0a06-457a-9ea1-efbf6fffba5c",
            "room_id": "11",
            "user_id": "2571",
            "username": "user2571",
            "message": "Test message 22570",
            "message_type": "TEXT",
            "server_id": "server-v2",
            "client_ip": "172.31.59.20",
            "created_at": "2026-03-30T03:17:05.490Z"
        },
        {
            "message_id": "7d72e48f-619f-439f-ae56-8e11ccfeacc5",
            "room_id": "11",
            "user_id": "2571",
            "username": "user2571",
            "message": "Test message 12570",
```

```
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T03:17:01.860Z"
            },
            {
                "message_id": "e92fe7ab-2058-44da-817b-2327be743f4a",
                "room_id": "11",
                "user_id": "2571",
                "username": "user2571",
                "message": "Test message 2570",
                "message_type": "TEXT",
                "server_id": "server-v2",
                "client_ip": "172.31.59.20",
                "created_at": "2026-03-30T03:16:57.146Z"
            }
        ]
    },
    "Q3_activeUsersInWindow": {
        "description": "Count active users in time window",
        "performanceTarget": "< 500 ms",
        "startTime": "2026-03-30T03:16:55.693839Z",
        "endTime": "2026-03-30T03:17:15.779312Z",
        "uniqueUserCount": 10000
    },
    "Q4_roomsForUser": {
        "description": "Get rooms user has participated in",
        "performanceTarget": "< 50 ms",
        "userId": "2571",
        "rooms": [
            {
                "room_id": "11",
                "last_activity": "2026-03-30T03:17:13.128Z",
                "message_count": 5
            }
        ]
    }
},
"analyticsQueries": {
    "A1_messagesPerTimeUnit": {
        "description": "Messages per second/minute statistics",
        "perMinute": [
            {
                "bucket": "2026-03-30T03:16:00.000Z",
                "message_count": 7982
            },
            {
                "bucket": "2026-03-30T03:17:00.000Z",
                "message_count": 42018
            }
        ],
        "perSecond": [
            {
                "bucket": "2026-03-30T03:16:55.000Z",
                "message_count": 395
            },
            {
                "bucket": "2026-03-30T03:16:56.000Z",
                "message_count": 1869
            },
            {
                "bucket": "2026-03-30T03:16:57.000Z",
                "message_count": 1696
            },
            {
                "bucket": "2026-03-30T03:16:58.000Z",
                "message_count": 1879
            },
            {
                "bucket": "2026-03-30T03:16:59.000Z",
                "message_count": 2143
            },
            {
                "bucket": "2026-03-30T03:17:00.000Z",
                "message_count": 2318
            },
            {
                "bucket": "2026-03-30T03:17:01.000Z",
                "message_count": 2779
            },
            {
```

```
                "bucket": "2026-03-30T03:17:02.000Z",
                "message_count": 2916
            },
            {
                "bucket": "2026-03-30T03:17:03.000Z",
                "message_count": 2734
            },
            {
                "bucket": "2026-03-30T03:17:04.000Z",
                "message_count": 2568
            },
            {
                "bucket": "2026-03-30T03:17:05.000Z",
                "message_count": 2572
            },
            {
                "bucket": "2026-03-30T03:17:06.000Z",
                "message_count": 2733
            },
            {
                "bucket": "2026-03-30T03:17:07.000Z",
                "message_count": 2508
            },
            {
                "bucket": "2026-03-30T03:17:08.000Z",
                "message_count": 2484
            },
            {
                "bucket": "2026-03-30T03:17:09.000Z",
                "message_count": 2576
            },
            {
                "bucket": "2026-03-30T03:17:10.000Z",
                "message_count": 2649
            },
            {
                "bucket": "2026-03-30T03:17:11.000Z",
                "message_count": 2726
            },
            {
                "bucket": "2026-03-30T03:17:12.000Z",
                "message_count": 2672
            },
            {
                "bucket": "2026-03-30T03:17:13.000Z",
                "message_count": 2878
            },
            {
                "bucket": "2026-03-30T03:17:14.000Z",
                "message_count": 2764
            },
            {
                "bucket": "2026-03-30T03:17:15.000Z",
                "message_count": 2141
            }
        ]
    },
    "A2_topActiveUsers": {
        "description": "Top 10 most active users",
        "topN": 10,
        "users": [
            {
                "user_id": "3542",
                "username": "user3542",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.507Z"
            },
            {
                "user_id": "3226",
                "username": "user3226",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.375Z"
            },
            {
                "user_id": "247",
                "username": "user247",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:12.260Z"
            },
            {
                "user_id": "2275",
```

```
                "username": "user2275",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.020Z"
            },
            {
                "user_id": "3202",
                "username": "user3202",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.364Z"
            },
            {
                "user_id": "8362",
                "username": "user8362",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:15.192Z"
            },
            {
                "user_id": "1822",
                "username": "user1822",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:12.844Z"
            },
            {
                "user_id": "3781",
                "username": "user3781",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.609Z"
            },
            {
                "user_id": "8902",
                "username": "user8902",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:15.394Z"
            },
            {
                "user_id": "2571",
                "username": "user2571",
                "message_count": 5,
                "last_active": "2026-03-30T03:17:13.128Z"
            }
        ]
    },
    "A3_topActiveRooms": {
        "description": "Top 10 most active rooms",
        "topN": 10,
        "rooms": [
            {
                "room_id": "1",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.771Z"
            },
            {
                "room_id": "9",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.775Z"
            },
            {
                "room_id": "4",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.773Z"
            },
            {
                "room_id": "8",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.774Z"
            },
            {
                "room_id": "12",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.776Z"
            },
            {
                "room_id": "10",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.775Z"
            },
            {
                "room_id": "7",
                "message_count": 2500,
                "last_activity": "2026-03-30T03:17:15.774Z"
            },
```

```
                {
                    "room_id": "13",
                    "message_count": 2500,
                    "last_activity": "2026-03-30T03:17:15.777Z"
                },
                {
                    "room_id": "17",
                    "message_count": 2500,
                    "last_activity": "2026-03-30T03:17:15.778Z"
                },
                {
                    "room_id": "15",
                    "message_count": 2500,
                    "last_activity": "2026-03-30T03:17:15.778Z"
                }
            ]
        },
        "A4_userParticipationPatterns": {
            "description": "User participation patterns (rooms visited distribution)",
            "distribution": [
                {
                    "rooms_visited": 1,
                    "user_count": 10000
                }
            ]
        }
    }
}
ubuntu@ip-172-31-59-20:~$
```

Write-Behind Proof Screeshots (

Right Screenshot: Published 10000/50000... (messages sent to RabbitMQ)

Left Screenshot: DB count still low

That gap is my write-behind proof)

```
Every 1.0s: psql -h 127.0.0.1 -U chatuser -d chatflow...  ip-172-31-59-20: Mon Mar 30 04:12:34 2026

 count
-------
     0
(1 row)
```

```
conn.close()
print("Done publishing 50000 messages")
EOF
Published 0/50000...
Published 5000/50000...
Published 10000/50000...
Published 15000/50000...
Published 20000/50000...
Published 25000/50000...
Published 30000/50000...
Published 35000/50000...
Published 40000/50000...
Published 45000/50000...
Done publishing 50000 messages
ubuntu@ip-172-31-59-20:~$ python3 << 'EOF'
import pika, json, uuid, time
from datetime import datetime, timezone

creds = pika.PlainCredentials('chatuser', 'chatpassword')
conn = pika.BlockingConnection(pika.ConnectionParameters('172.31.59.239', 5672, '/', cr
eds))
ch = conn.channel()

total = 50000
for i in range(total):
    room = str((i % 20) + 1)
    msg = {
        "messageId": str(uuid.uuid4()),
        "roomId": room,
        "userId": str(i % 10000 + 1),
        "username": f"user{i % 10000 + 1}",
        "message": f"Test message {i}",
        "messageType": "TEXT",
        "serverId": "server-v2",
        "clientIp": "172.31.59.20",
        "timestamp": datetime.now(timezone.utc).isoformat()
    }
    ch.basic_publish(exchange='chat.exchange', routing_key=f'room.{room}', body=json.du
mps(msg))
    if i % 5000 == 0:
        print(f"Published {i}/{total}...")
        time.sleep(0.5)

conn.close()
print("Done publishing 50000 messages")
EOF
Published 0/50000...
Published 5000/50000...
Published 10000/50000...
```

```
 count
-------
 44034
(1 row)
```

```
Published 20000/50000...
Published 25000/50000...
Published 30000/50000...
Published 35000/50000...
Published 40000/50000...
Published 45000/50000...
Done publishing 50000 messages
ubuntu@ip-172-31-59-20:~$ python3 << 'EOF'
import pika, json, uuid, time
from datetime import datetime, timezone

creds = pika.PlainCredentials('chatuser', 'chatpassword')
conn = pika.BlockingConnection(pika.ConnectionParameters('172.31.59.239', 5672, '/', cr
eds))
ch = conn.channel()

total = 50000
for i in range(total):
    room = str((i % 20) + 1)
    msg = {
        "messageId": str(uuid.uuid4()),
        "roomId": room,
        "userId": str(i % 10000 + 1),
        "username": f"user{i % 10000 + 1}",
        "message": f"Test message {i}",
        "messageType": "TEXT",
        "serverId": "server-v2",
        "clientIp": "172.31.59.20",
        "timestamp": datetime.now(timezone.utc).isoformat()
    }
    ch.basic_publish(exchange='chat.exchange', routing_key=f'room.{room}', body=json.du
mps(msg))
    if i % 5000 == 0:
        print(f"Published {i}/{total}...")
        time.sleep(0.5)

conn.close()
print("Done publishing 50000 messages")
EOF
Published 0/50000...
Published 5000/50000...
Published 10000/50000...
Published 15000/50000...
Published 20000/50000...
Published 25000/50000...
Published 30000/50000...
Published 35000/50000...
Published 40000/50000...
Published 45000/50000...
```

```
ubuntu@ip-172-31-59-20: ~

Every 1.0s: psql -h 127.0.0.1 -U chatuser -d chatflow...  ip-172-31-59-20: Mon Mar 30 04:07:03 2026

 count
-------
 50000
(1 row)
```

```
ubuntu@ip-172-31-59-20: ~

        time.sleep(0.5)

conn.close()
print("Done publishing 50000 messages")
EOF
^C
ubuntu@ip-172-31-59-20:~$
ubuntu@ip-172-31-59-20:~$ python3 << 'EOF'
import pika, json, uuid, time
from datetime import datetime, timezone

creds = pika.PlainCredentials('chatuser', 'chatpassword')
conn = pika.BlockingConnection(pika.ConnectionParameters('172.31.59.239', 5672, '/', cr
eds))
ch = conn.channel()

total = 50000
for i in range(total):
    room = str((i % 20) + 1)
    msg = {
        "messageId": str(uuid.uuid4()),
        "roomId": room,
        "userId": str(i % 10000 + 1),
        "username": f"user{i % 10000 + 1}",
        "message": f"Test message {i}",
        "messageType": "TEXT",
        "serverId": "server-v2",
        "clientIp": "172.31.59.20",
        "timestamp": datetime.now(timezone.utc).isoformat()
    }
    ch.basic_publish(exchange='chat.exchange', routing_key=f'room.{room}', body=json.du
mps(msg))
    if i % 5000 == 0:
        print(f"Published {i}/{total}...")
        time.sleep(0.5)

conn.close()
print("Done publishing 50000 messages")
EOF
Published 0/50000...
Published 5000/50000...
Published 10000/50000...
Published 15000/50000...
Published 20000/50000...
Published 25000/50000...
Published 30000/50000...
Published 35000/50000...
Published 40000/50000...
Published 45000/50000...
Done publishing 50000 messages
ubuntu@ip-172-31-59-20:~$
```

Data Integrity Check Screenshots:

```
ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "
SELECT COUNT(*) AS total_messages,
       COUNT(DISTINCT message_id) AS unique_message_ids,
       COUNT(*) - COUNT(DISTINCT message_id) AS duplicates
FROM messages;"
 total_messages | unique_message_ids | duplicates
----------------+--------------------+------------
          50000 |              50000 |          0
(1 row)
```

Circuit Breaker for database failure Screenshots: (

1. 6 messages that failed with "Failed to obtain JDBC Connection" — confirming PostgreSQL was unreachable

2. All failed at the same timestamp (04:31:50) — confirming it was a single circuit breaker event

3. Messages from different rooms (8-13) — showing the batch writer was mid-batch when it failed)

```
Every 2.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c 'SELECT COUNT(*) FROM messag...  ip-172-31-59-20: Mon Mar 30 04:33:06 2026

 count
-------
     6
(1 row)
```

```
ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "SELECT message_id, room_id, error_reason, failed_at FROM me
ssages_dlq LIMIT 10;"
              message_id              | room_id |           error_reason           |           failed_at
--------------------------------------+---------+----------------------------------+-----------------------------
 832a8c76-5afd-4fe2-b201-81d46712430e | 8       | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.133425+00
 0813886b-ad80-463f-b0ab-deaf7a5c3796 | 9       | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.139558+00
 1f8dd76c-c7a7-4e53-b255-8e6457c7e11b | 10      | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.141272+00
 a3f998ee-2135-4572-99b1-a9ec470059a5 | 11      | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.14319+00
 fe56e08d-bca3-49ae-a249-02e4a456eaa9 | 12      | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.14479+00
 29f964e6-57a4-434e-808d-3a28d0ca0507 | 13      | Failed to obtain JDBC Connection | 2026-03-30 04:31:50.539979+00
(6 rows)
```

Part 3 Screenshots:

Test 1 — Query Performance (Q1-Q4 response times)

```
ubuntu@ip-172-31-59-20:~$ for i in 1 2 3; do
  echo "Run $i:"
  time curl -s http://localhost:8082/api/metrics | python3 -c "
import sys,json
r=json.load(sys.stdin)
print('Q1 messageCount:', r['coreQueries']['Q1_roomMessagesInTimeRange']['messageCount'])
print('Q3 uniqueUsers:', r['coreQueries']['Q3_activeUsersInWindow']['uniqueUserCount'])
print('totalInDB:', r['totalMessagesInDB'])
"
  echo "---"
done
Run 1:
Q1 messageCount: 5500
Q3 uniqueUsers: 10000
totalInDB: 109987

real    0m0.605s
user    0m0.034s
sys     0m0.014s
---
Run 2:
Q1 messageCount: 5500
Q3 uniqueUsers: 10000
totalInDB: 109987

real    0m0.051s
user    0m0.035s
sys     0m0.012s
---
Run 3:
Q1 messageCount: 5500
Q3 uniqueUsers: 10000
totalInDB: 109987

real    0m0.056s
user    0m0.035s
sys     0m0.013s
---
```

Run	Response Time	What happened
Run 1	**605ms**	Cache miss — hit PostgreSQL, ran all queries
Run 2	**51ms**	Cache hit — returned cached response instantly
Run 3	**56ms**	Cache hit — same cached response

12x speedup from the response cache (605ms → 51ms). This directly demonstrates the Part 3 optimization: RESPONSE_CACHE_TTL_MS = 30_000 in [MetricsApiController.java](Assignment 3/consumer/src/main/java/org/example/chatflow/consumer/api/MetricsApiController.java)

Test 2 — Materialized Views populated

```
ubuntu@ip-172-31-59-20:~$ curl -s "http://localhost:8082/api/metrics" | python3 -c "
import sys,json
r=json.load(sys.stdin)
print('perMinute rows:', len(r['analyticsQueries']['A1_messagesPerTimeUnit']['perMinute']))
print('topUsers rows:', len(r['analyticsQueries']['A2_topActiveUsers']['users']))
print('topRooms rows:', len(r['analyticsQueries']['A3_topActiveRooms']['rooms']))
print('participation rows:', len(r['analyticsQueries']['A4_userParticipationPatterns']['distribution']))
"
perMinute rows: 4
topUsers rows: 10
topRooms rows: 10
participation rows: 1
```

Analytics Query	Rows	What it shows
A1 perMinute	**4**	Messages bucketed by minute — materialized view working
A2 topUsers	**10**	Top 10 most active users — materialized view working
A3 topRooms	**10**	Top 10 most active rooms — materialized view working
A4 participation	**1**	User participation distribution — materialized view working

All 4 analytics materialized views are populated and returning data. This demonstrates the Part 3 optimization: instead of running expensive aggregation queries on every API call, results are pre-computed in materialized views and refreshed every 30 seconds by StatsAggregator.

Optimization	Evidence
Indexing (idx_room_time, idx_user_time, idx_time)	Q1-Q4 meeting performance targets
HikariCP connection pooling	Pool visible in startup logs
Prepared statements	JdbcTemplate used throughout
Response cache	605ms → 51ms (12x speedup)
Materialized views	A1-A4 all returning data
Circuit breaker	DLQ entries when PostgreSQL down

Part 4 Testing

Test 1 — Baseline (500k messages)

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
          33454 |                  6 |        232127 |          99.90
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | ...  ip-172-31-59-20: Mon Mar 30 04:51:25 2026

Total queue depth: 0
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 04:51:44 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         165238 |                  9 |        280070 |          99.89
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | ...  ip-172-31-59-20: Mon Mar 30 04:52:02 2026

Total queue depth: 7
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 04:52:26 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         345446 |                  9 |        333508 |          99.88
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | ...  ip-172-31-59-20: Mon Mar 30 04:52:42 2026

Total queue depth: 4
```

```
ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "SELECT COUNT(*) FROM messages;"
 count
--------
 500000
(1 row)

ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "
SELECT
    numbackends AS active_connections,
    xact_commit AS total_commits,
    ROUND(blks_hit::numeric / NULLIF(blks_hit+blks_read,0)*100,2) AS buffer_hit_pct,
    deadlocks,
    conflicts
FROM pg_stat_database WHERE datname='chatflow';"
 active_connections | total_commits | buffer_hit_pct | deadlocks | conflicts
--------------------+---------------+----------------+-----------+-----------
                 10 |        379196 |          99.88 |         0 |         0
(1 row)
```

Test 1 results summary:

Metric	Value
Total messages written	**500,000**
Publish rate	**3,914 msg/s**
Active connections	**10**
Buffer hit rate	**99.88%**
Deadlocks	**0**
Queue depth (peak)	**7**

Test 2 — Stress Test (1,000,000 messages)

```
ubuntu@ip-172-31-59-20:~$ python3 << 'EOF'
import pika, json, uuid, time
from datetime import datetime, timezone

creds = pika.PlainCredentials('chatuser', 'chatpassword')
conn = pika.BlockingConnection(pika.ConnectionParameters('172.31.59.239', 5672, '/', creds))
ch = conn.channel()

total = 1000000
start = time.time()
for i in range(total):
    room = str((i % 20) + 1)
    msg = {
        "messageId": str(uuid.uuid4()),
        "roomId": room,
        "userId": str(i % 10000 + 1),
        "username": f"user{i % 10000 + 1}",
        "message": f"Test message {i}",
        "messageType": "TEXT",
        "serverId": "server-v2",
        "clientIp": "172.31.59.20",
        "timestamp": datetime.now(timezone.utc).isoformat()
    }
    ch.basic_publish(exchange='chat.exchange', routing_key=f'room.{room}', body=json.dumps(msg))
    if i % 100000 == 0 and i > 0:
        elapsed = time.time() - start
        print(f"Published {i}/{total} | Rate: {i/elapsed:.0f} msg/s | Elapsed: {elapsed:.1f}s")

elapsed = time.time() - start
conn.close()
print(f"\nDone! Published {total} messages in {elapsed:.1f}s")
print(f"Publish rate: {total/elapsed:.0f} msg/s")
EOF

Published 100000/1000000 | Rate: 2728 msg/s | Elapsed: 36.7s
Published 200000/1000000 | Rate: 3296 msg/s | Elapsed: 60.7s
Published 300000/1000000 | Rate: 3523 msg/s | Elapsed: 85.2s
Published 400000/1000000 | Rate: 3662 msg/s | Elapsed: 109.2s
Published 500000/1000000 | Rate: 3780 msg/s | Elapsed: 132.3s
Published 600000/1000000 | Rate: 3846 msg/s | Elapsed: 156.0s
Published 700000/1000000 | Rate: 3896 msg/s | Elapsed: 179.7s
Published 800000/1000000 | Rate: 3947 msg/s | Elapsed: 202.7s
Published 900000/1000000 | Rate: 3935 msg/s | Elapsed: 228.7s

Done! Published 1000000 messages in 252.5s
Publish rate: 3960 msg/s
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 04:59:27 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
          30564 |                  9 |        380744 |          99.89
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | ...  ip-172-31-59-20: Mon Mar 30 04:59:46 2026

Total queue depth: 0
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 04:59:58 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         132562 |                  9 |        421645 |          99.89
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | ...  ip-172-31-59-20: Mon Mar 30 05:00:17 2026

Total queue depth: 4
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         699051 |                 10 |        588071 |          99.87
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | python3 -c "import sys,json; qs=json.load(sys.stdin); total=sum(q.get('message...  ip-172-31-59-20: Mon Mar 30 05:02:30 2026

Total queue depth: 1
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         815345 |                 10 |        622200 |          99.86
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | python3 -c "import sys,json; qs=json.load(sys.stdin); total=sum(q.get('message...  ip-172-31-59-20: Mon Mar 30 05:03:01 2026

Total queue depth: 4
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
        1000000 |                 11 |        677067 |          99.86
(1 row)
```

```
Every 3.0s: curl -s -u chatuser:chatpassword http://172.31.59.239:15672/api/queues | python3 -c "import sys,json; qs=json.load(sys.stdin); total=sum(q.get('message...  ip-172-31-59-20: Mon Mar 30 05:03:47 2026

Total queue depth: 0
```

```
ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "SELECT COUNT(*) FROM messages;"
psql -h 127.0.0.1 -U chatuser -d chatflow -c "
SELECT
    numbackends AS active_connections,
    xact_commit AS total_commits,
    ROUND(blks_hit::numeric / NULLIF(blks_hit+blks_read,0)*100,2) AS buffer_hit_pct,
    deadlocks,
    conflicts
FROM pg_stat_database WHERE datname='chatflow';"
  count
---------
 1000000
(1 row)

 active_connections | total_commits | buffer_hit_pct | deadlocks | conflicts
--------------------+---------------+----------------+-----------+-----------
                  6 |        680194 |          99.92 |         0 |         0
(1 row)
```

Key observations:

- **No bottleneck** — queue depth stayed near 0, consumer kept up
- **Buffer hit rate stable** at 99.87-99.89% — no memory pressure
- **Connections stable** at 9-10 — no pool exhaustion
- **Publish rate stable** at ~3,960 msg/s — no degradation curve

<u>Test 3 — Endurance Test (30 minutes at 80% throughput).</u>

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 05:34:32 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
          12563 |                  6 |        684464 |          99.93
(1 row)
```

```
Every 30.0s: echo '=== Memory ===' && free -h && echo '=== Disk ===' && df -h /var/li...  ip-172-31-59-20: Mon Mar 30 05:34:28 2026

=== Memory ===
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       442Mi       570Mi       436Mi       896Mi       854Mi
Swap:             0B          0B          0B
=== Disk ===
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       7.6G  4.5G  3.1G  60% /
=== DB Connections ===
 count
-------
     6
(1 row)
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 06:01:50 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
         2876400 |                 11 |       1458403 |          99.53
(1 row)
```

```
Every 30.0s: echo '=== Memory ===' && free -h && echo '=== Disk ===' && df -h /var/li...  ip-172-31-59-20: Mon Mar 30 06:02:05 2026

=== Memory ===
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       961Mi        76Mi       599Mi       872Mi       174Mi
Swap:             0B          0B          0B
=== Disk ===
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       7.6G  5.8G  1.9G  76% /
=== DB Connections ===
 count
-------
    16
(1 row)
```

```
Every 5.0s: psql -h 127.0.0.1 -U chatuser -d chatflow -c "                    ip-172-31-59-20: Mon Mar 30 06:15:23 2026

 total_messages | active_connections | total_commits | buffer_hit_pct
----------------+--------------------+---------------+----------------
        3123900 |                  8 |       1512706 |          99.29
(1 row)
```

```
Every 30.0s: echo '=== Memory ===' && free -h && echo '=== Disk ===' && df -h /var/li...  ip-172-31-59-20: Mon Mar 30 06:15:38 2026

=== Memory ===
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi       988Mi       129Mi       599Mi       792Mi       156Mi
Swap:             0B          0B          0B
=== Disk ===
Filesystem      Size  Used Avail Use% Mounted on
/dev/root       7.6G  6.3G  1.4G  83% /
=== DB Connections ===
 count
-------
     7
(1 row)
```

```
ubuntu@ip-172-31-59-20:~$ python3 << 'EOF'
import pika, json, uuid, time
from datetime import datetime, timezone

creds = pika.PlainCredentials('chatuser', 'chatpassword')
conn = pika.BlockingConnection(pika.ConnectionParameters('172.31.59.239', 5672, '/', creds))
ch = conn.channel()

target_rate = 3168  # 80% of 3960 msg/s
duration_secs = 1800  # 30 minutes
batch = 100  # publish in batches of 100
sleep_per_batch = batch / target_rate  # seconds between batches

start = time.time()
total_published = 0
report_interval = 300  # report every 5 minutes

print(f"Starting endurance test: target={target_rate} msg/s for {duration_secs//60} minutes")

while time.time() - start < duration_secs:
    for _ in range(batch):
        i = total_published
        room = str((i % 20) + 1)
        msg = {
            "messageId": str(uuid.uuid4()),
            "roomId": room,
            "userId": str(i % 10000 + 1),
            "username": f"user{i % 10000 + 1}",
            "message": f"Test message {i}",
            "messageType": "TEXT",
            "serverId": "server-v2",
            "clientIp": "172.31.59.20",
            "timestamp": datetime.now(timezone.utc).isoformat()
        }
        ch.basic_publish(exchange='chat.exchange', routing_key=f'room.{room}', body=json.dumps(msg))
        total_published += 1

    elapsed = time.time() - start
EOFnt(f"Average rate: {total_published/elapsed:.0f} msg/s")blished:,} | Actual rate: {actual_rate:.0f} msg/s")
Starting endurance test: target=3168 msg/s for 30 minutes
[0.0 min] Published: 100 | Actual rate: 5318 msg/s
[0.0 min] Published: 200 | Actual rate: 2148 msg/s
[0.0 min] Published: 300 | Actual rate: 1830 msg/s
[0.0 min] Published: 400 | Actual rate: 1733 msg/s
[0.0 min] Published: 500 | Actual rate: 1642 msg/s
[0.0 min] Published: 600 | Actual rate: 1658 msg/s
[0.0 min] Published: 700 | Actual rate: 1651 msg/s
[0.0 min] Published: 800 | Actual rate: 1637 msg/s
[0.0 min] Published: 900 | Actual rate: 1549 msg/s
[0.0 min] Published: 1,000 | Actual rate: 1554 msg/s
[0.0 min] Published: 1,100 | Actual rate: 1522 msg/s
[0.0 min] Published: 1,200 | Actual rate: 1452 msg/s
[0.0 min] Published: 1,300 | Actual rate: 1420 msg/s
[5.0 min] Published: 530,400 | Actual rate: 1768 msg/s
[5.0 min] Published: 530,500 | Actual rate: 1768 msg/s
[5.0 min] Published: 530,600 | Actual rate: 1768 msg/s
[5.0 min] Published: 530,700 | Actual rate: 1768 msg/s
[5.0 min] Published: 530,800 | Actual rate: 1768 msg/s
[5.0 min] Published: 530,900 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,000 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,100 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,200 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,300 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,400 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,500 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,600 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,700 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,800 | Actual rate: 1768 msg/s
[5.0 min] Published: 531,900 | Actual rate: 1768 msg/s
[5.0 min] Published: 532,000 | Actual rate: 1768 msg/s
```

```
[10.0 min] Published: 1,066,400 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,066,500 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,066,600 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,066,700 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,066,800 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,066,900 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,000 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,100 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,200 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,300 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,400 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,500 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,600 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,700 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,800 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,067,900 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,068,000 | Actual rate: 1777 msg/s
[10.0 min] Published: 1,068,100 | Actual rate: 1777 msg/s
[15.0 min] Published: 1,595,400 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,595,500 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,595,600 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,595,700 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,595,800 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,595,900 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,000 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,100 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,200 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,300 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,400 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,500 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,600 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,700 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,800 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,596,900 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,597,000 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,597,100 | Actual rate: 1773 msg/s
[15.0 min] Published: 1,597,200 | Actual rate: 1773 msg/s
[20.0 min] Published: 2,118,400 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,118,500 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,118,600 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,118,700 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,118,800 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,118,900 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,000 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,100 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,200 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,300 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,400 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,500 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,600 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,700 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,800 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,119,900 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,120,000 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,120,100 | Actual rate: 1765 msg/s
[20.0 min] Published: 2,120,200 | Actual rate: 1765 msg/s
[25.0 min] Published: 2,632,200 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,300 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,400 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,500 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,600 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,700 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,800 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,632,900 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,633,000 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,633,100 | Actual rate: 1755 msg/s
[25.0 min] Published: 2,633,200 | Actual rate: 1755 msg/s
```

```
Endurance test complete!
Duration: 30.0 minutes
Total published: 3,123,900
Average rate: 1735 msg/s
ubuntu@ip-172-31-59-20:~$
```

```
ubuntu@ip-172-31-59-20:~$ psql -h 127.0.0.1 -U chatuser -d chatflow -c "SELECT COUNT(*) FROM messages;"
psql -h 127.0.0.1 -U chatuser -d chatflow -c "
SELECT numbackends AS active_connections,
       ROUND(blks_hit::numeric / NULLIF(blks_hit+blks_read,0)*100,2) AS buffer_hit_pct,
       deadlocks FROM pg_stat_database WHERE datname='chatflow';"
  count
---------
 3123900
(1 row)

 active_connections | buffer_hit_pct | deadlocks
--------------------+----------------+-----------
                  7 |          99.31 |         0
(1 row)
```

Throughput stability (no degradation):

Time	Published	Actual Rate
5 min	530,400	1,768 msg/s
10 min	1,067,000	1,777 msg/s
15 min	1,596,600	1,773 msg/s
20 min	2,119,000	1,765 msg/s
25 min	2,633,000	1,755 msg/s
30 min	3,123,900	1,735 msg/s

Rate stayed within 2.5% of opening rate — no performance degradation.

Resource progression:

Time	Memory Used	Disk Used	DB Connections	Buffer Hit
Start	442MB	60%	6	99.93%
15 min	961MB	76%	16	99.53%
30 min	988MB	83%	7	99.29%

Key findings:

1. No memory leak — memory stabilized at ~988MB after 15 min, did not keep growing
2. No connection pool exhaustion — connections peaked at 16 (under HikariCP max of 20), settled back to 7

3. Disk is the bottleneck — grew from 60% to 83% over 30 min (~3.1M messages). At this rate, disk would fill in ~2 hours of sustained load. This is the primary scaling concern to document
4. Buffer hit rate dropped slightly (99.93% → 99.29%) as dataset grew, still excellent.